Lehrstuhl für Technische Mechanik

Bridging Interphase and Interface Models for Composites – Parameter Identification in Mechanical Problems

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Declaration of Authorship

I hereby declare that I have written this Master's Thesis independently, unaided and without using sources other than those listed. All content that was taken from a third party, either verbatim or in substance, have been acknowledged.

This thesis has never been submitted to any other examination authority and never been accepted as part of an examination achievement, neither in this form nor in a similar form.

I also declare that I used AI for assistance with grammar correction, and enhancing the readability of the text. This tool was used solely to enhance the clarity and accuracy of my work and did not contribute to the ideas presented in this thesis.

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Abstract

The mechanical behaviour of composite materials depends strongly on the thin interphase region between inclusion and matrix. Explicitly modelling this region in numerical simulations is expensive, so it is often replaced by a zero-thickness interface characterised by a small set of phenomenological parameters. However, identifying these parameters from known interphase properties is a non-trivial inverse problem.

In this work, we develop a physics-informed neural network (PINN) that identifies the four parameters of the Extended General Interface Model directly from target effective properties computed via a three-layer interphase model. The network is trained on Mori-Tanaka homogenization data and embeds the governing equations into its loss function to ensure physically self-consistent predictions.

We apply the framework to two composite geometries. For particle-reinforced composites (CSA), the inverse PINN reconstructs effective properties with errors below 2% for interpolated coating profiles, and a structured sweep over 36 stiffness ratio to volume fraction combinations confirms that 33 pass the 5% prediction threshold. For fibre-reinforced composites (CCA), the transverse shear response cannot be matched within the interface framework: even a per-configuration optimiser leaves a persistent G_{tr} gap of approximately 23%, pointing to a fundamental limitation of the zero-thickness approximation for fibre composites.

Keywords: Composite Materials, Interphase, Extended General Interface Model, Physics-Informed Neural Networks, Inverse Parameter Identification, Mori-Tanaka Homogenization, CCA, CSA.

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Notations

Continuum Mechanics:	$\boldsymbol{u}$	Displacement vector
	$\boldsymbol{\sigma}$	Cauchy stress tensor
	$\boldsymbol{\varepsilon}$	Strain tensor
	$\boldsymbol{t}$	Surface traction vector
	$\boldsymbol{n}$, $\bar{\boldsymbol{n}}$	Outward unit normal
	$\mathbb{C}$	Fourth-order stiffness tensor
	λ, μ	Lamé parameters
	κ	Bulk modulus (plane-strain: $\lambda+\mu$; 3D: $\lambda+\frac{2}{3}\mu$)
	E, ν	Young's modulus, Poisson's ratio
	$\boldsymbol{I}$	Second-order identity tensor
	$\mathbb{I}$	Fourth-order identity tensor

Geometry and Phases:	$\mathcal{V}$	RVE volume
	$\mathcal{B}$	Bulk domain
	$\mathcal{I}$	Interface surface
	N	Number of coating layers
	r_m	Outer radius of phase m
	f	Inclusion volume fraction
	c_m	Sub-phase volume fraction
	ϕ_m	Volume fraction ratio

Concentration Tensors:	$\mathbb{T}^{(q)}$	Strain concentration tensor of phase q
	$\mathbb{H}^{(q)}$	Stress-strain concentration tensor of phase q
	$\mathbb{T}^{\mathrm{eq}}$, $\mathbb{H}^{\mathrm{eq}}$	Equivalent concentration tensors
	T_κ, T_{44}, T_{55}	Independent scalar components of $\mathbb{T}$ (CCA)
	T_μ	Independent shear component of $\mathbb{T}$ (CSA)

Transformations:	$\boldsymbol{R}$	Rotation matrix
	$\boldsymbol{Q}_\varepsilon$, $\boldsymbol{Q}_\sigma$	Voigt transformation matrices for strain and stress

	$[\![\cdot]\!]$	Jump operator across interface
	$\{\{\cdot\}\}$	Average operator across interface
	$\bar{k}$	Interface orthogonal stiffness
Interface / EGIM:	$\bar{\lambda}$	Interface Lamé parameter
	$\bar{\mu}$	Interface shear modulus
	$\bar{\alpha}$	Interface position parameter
	$\hat{\boldsymbol{\varepsilon}}$, $\hat{\boldsymbol{\sigma}}$	Interface contributions to macroscopic fields

	$\boldsymbol{\varepsilon}^{(0)}$	Applied far-field strain
	X_i	BVP amplitude coefficients
Homogenization:	${}^{\mathrm{M}}\mathbb{C}$	Macroscopic effective stiffness tensor
	$K_{\mathrm{tr}}, G_{\mathrm{tr}}, G_{\mathrm{ax}}$	Effective CCA moduli
	$K_{\mathrm{eff}}, G_{\mathrm{eff}}$	Effective CSA moduli (bulk, shear)

	$\boldsymbol{W}^{(\ell)}, \boldsymbol{b}^{(\ell)}$	Weight matrix and bias vector of layer ℓ
Neural Network:	$\boldsymbol{\theta}$	Trainable network parameters
	$\mathcal{L}$	Loss function

	(1)	Inclusion/core phase
	(0)	Far-field condition
	(2)	Matrix (EGIM)/first coating (interphase)
Superscripts/Subscripts:	(q)	General phase index
	eq	Equivalent inhomogeneity
	${}^{\mathrm{M}}$	Macroscopic quantity
	$\bar{(\cdot)}$	Interface quantity

Chapter 1

Introduction

Weight reduction has caused a huge shift in the materials used to build things over the past couple of decades especially in the aerospace industry. Carbon fiber reinforced polymers (CFRPs), a high performance composite material, are now part of the majority of airframe components in modern aircrafts like the Boeing 787 and Airbus A350. These planes use more than 50% composite materials by structural weight[1].

The appeal of composite materials lies in their exceptional strength-to-weight ratios that far exceed conventional metals. A unidirectional CFRP laminate can achieve tensile moduli exceeding 150 GPa at densities around 1.6 g/cm^3, compared to aluminum's 70 GPa at 2.7 g/cm^3 [2]. However, this performance is not an intrinsic property of the constituents alone. The overall behavior of a composite depends on its underlying microstructure specifically, on the orientation, distribution, volume fraction, and shape of its constituents, as well as the interactions between them [3]. Predicting this overall response from such complex microstructures is a challenging task that has motivated the development of various micromechanical methods, among which homogenization has proven particularly powerful.

1.1 Homogenization

Homogenization aims to estimate the macroscopic behavior of a heterogeneous material based on the response of its underlying microstructure, that can replace the heterogeneous material with an equivalent homogeneous medium [4]. The foundations were laid by variational bounds from Voigt-Reuss [5], [6] to the Hashin-Shtrikman bounds [7] and by Eshelby's work [8] on the elastic field of an ellipsoidal inclusion. Building on these, the Mori-Tanaka method [9], [10] provides explicit estimates for effective properties by assuming each inclusion experiences the average matrix strain as its far-field condition, while the composite cylinder assemblage (CCA) [11] offers exact solutions for specific fiber composite geometries.

A limitation of classical homogenization, however, is that it cannot capture size-dependent material behavior. Since classical formulations lack a physical length-scale, they cannot account for the pronounced surface and interface effects that arise at small scales for instance, in nanocomposites where the area-to-volume ratio becomes significant [12], [13]. To address this problem, one has to explicitly include the region where the different phases meet.

1.2 The Interphase

The region where fiber and matrix join is termed as the *interphase* which is a three-dimensional zone of finite thickness where material properties transition from those of the reinforcement to those of the matrix. This region forms through various mechanisms creating chemically distinct layers with unique properties typically with 50 to 500 nm as thickness [14]. These Interphases significantly influence overall composite behavior since load transfer from matrix to fiber occurs entirely through this region [15].

The primary approach to study interphases was to consider them as a homogeneous independent phase between constituents [16]. Luo and Weng [17], [18] developed a modified Mori-Tanaka method for composites with coated fibers and particles, while Benveniste et al. [19] obtained effective moduli for composites with coated fibers. For a more realistic analysis, interphases can be assumed to possess spatially variable properties like inhomogeneous, graded, or multilayered. Hervé and Zaoui [20], [21] derived elastic fields for multilayered inclusions, providing a framework suited to analyze graded interphases.

1.2.1 Computational Challenge

From a computational perspective, the interphase presents a significant challenge. Finite element analysis requires discretizing this thin layer, but the extreme aspect ratios involved a 100 nm interphase surrounding a 7 μm diameter carbon fiber lead to severely ill-conditioned stiffness matrices and prohibitive mesh densities [22]. For RVEs containing hundreds of fibers, explicitly resolving each interphase becomes computationally expensive. This tension between physical accuracy and computational practicality motivates the development of simplified mathematical descriptions which replaces the interphase with finite volume with a zero-thickness *interface* model.

1.3 Interface Models:

A well-established strategy to address the computational challenges of finite-thickness interphases is to replace them with zero-thickness *interface* models characterized by field discontinuities [23]. In elasticity, these discontinuities manifest as jumps in displacement and/or traction across the interface. This approach trades geometric fidelity for computational efficiency while preserving the essential mechanical response. Figure 1.1 illustrates the hierarchy of interface and interphase models studied in this thesis.

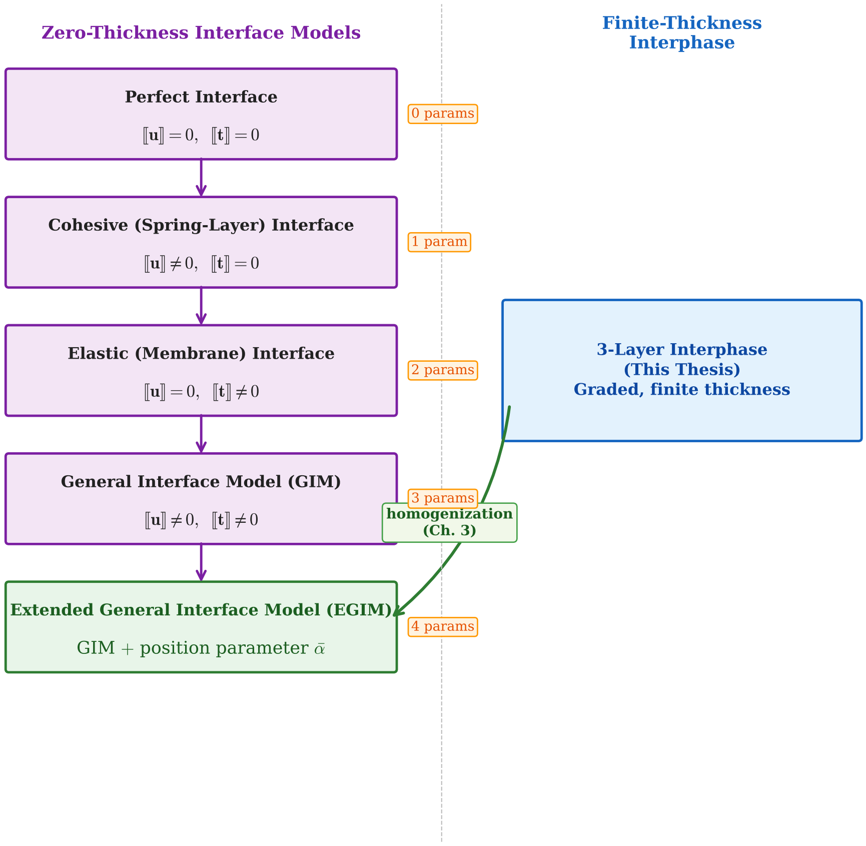

Figure 1.1: Hierarchy of interface and interphase models

Interface models form a hierarchy of increasing generality, as illustrated in the left branch of Figure 1.1. The simplest is the *perfect interface*, which enforces continuity of both displacement and traction across the surface convenient but unable to capture any interphase physics. Relaxing one continuity condition while maintaining the other leads to two classical imperfect models. The *cohesive* (spring-layer) interface [24], [25] allows displacement jumps. The *elastic* (membrane) interface [26], [27], permits traction jumps driven by surface stresses while maintaining displacement continuity, and captures stiff interphases that resist tangential deformation.

Real interphases often exhibit characteristics of both compliant and stiff behaviour simultaneously. Hashin [28] and Benveniste and Miloh [29] unified the above into a *general interface model* (GIM) that admits both displacement and traction jumps. All earlier models emerge as limiting cases of the GIM. However, the GIM assumes the interface lies at the geometric mid-plane of the original interphase, an assumption that becomes problematic for graded or asymmetric interphases. Firooz et al. [3], [22] addressed this by introducing the *Extended General Interface Model* (EGIM), which adds a position parameter $\bar{\alpha} \in [0, 1]$ that allows the interface to sit anywhere within the original interphase region.

The complete EGIM is thus characterised by four independent parameters:

- $\bar{k}$ — orthogonal stiffness
- $\bar{\lambda}$ — interface Lamé parameter
- $\bar{\mu}$ — interface shear modulus
- $\bar{\alpha}$ — position parameter

1.4 The Research Gap: Parameter Identification

The theoretical framework described in section above provides a path from finite-thickness interphases to computationally efficient interface models. However, a critical gap remains in practical application: the interface parameters ($\bar{k}$, $\bar{\lambda}$, $\bar{\mu}$, $\bar{\alpha}$) are phenomenological quantities that cannot be directly measured or straightforwardly derived from known interphase properties [3].

Consider an example: given a CFRP system with a sizing layer of known thickness h and elastic moduli (λ_{sizing}, μ_{sizing}), what interface stiffness $\bar{k}$ produces the same effective composite response? The answer is not obvious, and several complicating factors arise:

Non-uniqueness: The interface parameters interact in complex ways different combinations of $\bar{k}$ and $\bar{\alpha}$ can yield identical macroscopic responses [3], and a single parameter like $\bar{k}$ affects both bulk and shear moduli simultaneously, making independent identification from individual loading cases difficult [22].

Graded Properties: Real interphases rarely exhibit uniform properties. Sizing agents create cross-linking density gradients [30], diffusion processes generate compositional profiles, and damage accumulation produces spatially varying stiffness [14], [31]. Mapping such complexity onto equivalent interface parameters requires systematic methodology.

1.4.1 Approaches to Parameter Identification

Two fundamentally different strategies exist for determining interface parameters from interphase descriptions: asymptotic derivation and direct parametric identification. Understanding their limitations motivates the data-driven approach adopted in this thesis.

Asymptotic Approach. Asymptotic analysis treats the interphase thickness ε as a small parameter and examines the limiting behavior as $\varepsilon \to 0$. This approach is pioneered by Bövik [32] and developed by Benveniste [33]. This methodology expands the displacement and stress fields within the interphase as Taylor series about the mid-surface, then enforces continuity conditions at the boundaries. The *scaling* of interphase stiffness with thickness determines the resulting interface type: soft scaling ($\mathbb{C} \sim \varepsilon$) yields a cohesive interface with $\bar{k} = \mu/h$, while stiff scaling ($\mathbb{C} \sim \varepsilon^{-1}$) produces a membrane-type interface [34], [35]. Although it looks elegant, this approach assumes comprehensive understanding of the interphase properties and related scaling regime conditions, which are rarely met for real interphases that are spatially graded or only partially characterised through indirect measurements[34].

Direct Parametric Identification. An alternative strategy requires the interface model to reproduce the same effective composite response as the interphase model. One seeks interface parameters $(\bar{k}, \bar{\lambda}, \bar{\mu}, \bar{\alpha})$ such that:

$$\mathbb{T}^{\mathrm{eq}}_{\mathrm{interface}}(\bar{k}, \bar{\lambda}, \bar{\mu}, \bar{\alpha}) = \mathbb{T}^{\mathrm{eq}}_{\mathrm{interphase}} \tag{1.1}$$

This matching condition leads to an *underdetermined* system: the EGIM has four unknown parameters, but the concentration tensors provide at most two (CSA) or three (CCA) independent scalar equations [3]. Straightforward analytical inversion is therefore not possible, and the system must be solved numerically a task well suited to data-driven methods.

1.5 Bridging the Gap: A Data-Driven Approach

The limitations outlined above suggest that purely analytical routes to parameter identification have inherent constraints. This thesis develops a computational framework that combines analytical homogenization with neural networks to address these challenges.

We develop a physics-informed neural network (PINN) that directly identifies interface parameters from target mechanical response. Unlike iterative inverse optimisation, which requires repeated forward solves at each iteration, the trained PINN identifies parameters in a single forward pass; the physics loss embedded during training [36] ensures that predictions remain consistent with the governing homogenization equations. We apply the same inverse network to both CSA and CCA geometries to test the scope of the interface model across different composite types.

Neural networks are preferred over alternatives such as Gaussian process regression [37] or polynomial chaos expansions [38] because the mapping is highly nonlinear, the input space is seven- to eight-dimensional, and the training datasets contain around 200 000 samples(CCA) a regime where Gaussian processes become computationally prohibitive [39]. The detailed justification is given in Section 3.3.

1.6 Objectives

The primary objective of this thesis is to investigate the relationship between finite-thickness interphase models and zero-thickness interface models in composite homogenization, and to develop neural network approaches where analytical methods reach their limits. Specifically, the research aims to:

1. **Implement and Verify the Multi-Coated Interphase Model:** For both fiber-reinforced (CCA) and particle-reinforced (CSA) composites, establishing a trusted reference solution through consistency checks and limiting-case verification.

2. **Develop an Inverse PINN for Parameter Identification:** Construct a physics-informed neural network that identifies interface parameters from target mechanical response, embedding the homogenization equations directly into the training process.

3. **Apply to Both Composite Geometries:** Deploy the inverse PINN on CSA and CCA composites to recover Extended Interface parameters that reproduce the three-layered interphase effective properties, and examine where the method succeeds and where it encounters limits.

4. **Synthesise the Findings:** Establish when the Extended Interface model is applicable and when it is not, defining clear scope boundaries for different composite geometries.

1.7 Structure of the Thesis

This work is structured as follows:

Chapter 2 presents the theoretical foundations. It begins with the governing equations of continuum mechanics and homogenization which includes balance laws, the Hill-Mandel condition, and the Eshelby inclusion problem. The coordinate transformations for cylindrical and spherical geometries is then established. The multi-coated inclusion model for interphase analysis is developed, the concentration tensor structures are derived, and the interface models from perfect to extended general are formulated. The interface-enhanced Mori-Tanaka homogenization framework is developed, and the chapter concludes with an introduction to feedforward neural networks and physics-informed learning.

Chapter 3 develops the methodology. The boundary value problems for the multi-layered interphase model and the interface-enhanced Mori-Tanaka model are derived for fibre-reinforced (CCA) and particle-reinforced (CSA) composites. The neural network architecture for inverse parameter identification is presented, along with training procedures and data generation.

Chapter 4 presents the results and discussion. The three-layered interphase model is first verified through consistency checks and limiting cases. Inverse PINNs for both CSA and CCA geometries are demonstrated, with the CCA case providing independent confirmation of the gap. The limits of the Extended Interface Model are quantified through systematic gap analysis with numerical examples. The chapter concludes with a discussion of the findings and practical recommendations.

Chapter 5 summarises the contributions, discusses limitations of the current framework, and outlines directions for future research.

Chapter 2

Theoretical Foundations

This chapter presents the theoretical foundations of our present work. We begin with a brief set of fundamentals in Section 2.1 starting with the Representative Volume Element, balance of momentum, constitutive relations, and the Hill-Mandel and Eshelby formulations. Section 2.2 then develops the three-layered interphase model, where an interphase region is discretised into coating layers with piecewise-constant properties. We define the volume fractions, concentration tensors, and homogenization formulas for both fibre-reinforced (2D, CCA) and particle-reinforced (3D, CSA) composites. Section 2.3 presents the alternative approach: the Extended General Interface Model (EGIM), which replaces the finite-thickness interphase with a zero-thickness surface governed by interface constitutive laws. We develop the interface kinematics, model hierarchy, modified averaging relations, and the Mori-Tanaka homogenization with interface effects. Section 2.4 then introduces the neural network and physics-informed learning concepts needed for the inverse parameter Identification which we will continue in Chapter 3.

2.1 Preliminaries

Before setting up the two models, we establish the continuum mechanics framework that supports homogenization.

2.1.1 Representative Volume Element

Homogenization relies on the concept of a Representative Volume Element (RVE) which is basically a material domain $\mathcal{V}$ that is large enough to capture the characteristics of the microstructure yet small enough that macroscopic fields remain approximately uniform across it [4]. This separation of scales,

$$\ell_{\text{micro}} \ll \ell_{\text{RVE}} \ll \ell_{\text{macro}}, \tag{2.1}$$

where ℓ_{micro} is the characteristic inclusion size (fiber radius or particle diameter), ℓ_{RVE} is the RVE dimension, and ℓ_{macro} is the structural length scale. This condition is the basic requirement for any mean-field homogenization scheme. When this condition holds, the heterogeneous material can be replaced by an equivalent homogeneous medium whose effective stiffness ${}^{\mathrm{M}}\mathbb{C}$ relates macroscopic stress and strain.

For composite cylinder and composite sphere assemblages used in this work, the RVE is a single coated inclusion surrounded by matrix material. Hashin [11] showed that assemblages of such unit cells, packed to fill all available space, yield exact effective properties.

2.1.2 Balance of Linear Momentum

Within the RVE, each phase(fiber/coatings/matrix) satisfies the static balance of linear momentum in the absence of body forces:

$$\operatorname{Div} \boldsymbol{\sigma} = \mathbf{0}, \quad \text{in } \mathcal{B} \tag{2.2}$$

$$t = \boldsymbol{\sigma} \cdot n \quad \text{on } \partial B \tag{2.3}$$

where $\mathcal{B}$ denotes the bulk domain (all phases).

2.1.3 Constitutive Relations

Both the models in this thesis assume linear elastic, isotropic phase materials. The strain and stress tensors in Voigt notation are:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{22} & \varepsilon_{33} & 2\varepsilon_{12} & 2\varepsilon_{13} & 2\varepsilon_{23} \end{bmatrix}^T \tag{2.4}$$

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{22} & \sigma_{33} & \sigma_{12} & \sigma_{13} & \sigma_{23} \end{bmatrix}^T \tag{2.5}$$

For isotropic phases, the stiffness tensor takes the form:

$$\mathbb{C} = \lambda \boldsymbol{I} \otimes \boldsymbol{I} + 2\mu \, \mathbb{I}^{\mathrm{sym}} \tag{2.6}$$

where λ and μ are the Lamé parameters. In matrix form:

$$\mathbb{C} = \begin{bmatrix} \lambda + 2\mu & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2\mu & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{bmatrix} \tag{2.7}$$

The plane-strain bulk modulus is $\kappa = \lambda + \mu$ and the 3D bulk modulus is $\kappa = \lambda + \frac{2}{3}\mu$.

2.1.4 Hill-Mandel Condition

The Hill-Mandel condition ensures energy consistency between the micro and macro scales. For a body $\mathcal{B}$ with boundary $\partial\mathcal{B}$ subject to linear displacement boundary conditions $\boldsymbol{u} = {}^{\mathrm{M}}\boldsymbol{\varepsilon} \cdot \boldsymbol{x}$, the macroscopic strain and stress are defined as boundary integrals [40]:

$$^{\mathrm{M}}\boldsymbol{\varepsilon} = \frac{1}{\mathcal{V}} \int_{\partial\mathcal{B}} \frac{1}{2} \left(\boldsymbol{u} \otimes \boldsymbol{n} + \boldsymbol{n} \otimes \boldsymbol{u}\right) \mathrm{d}A \tag{2.8}$$

$$^{\mathrm{M}}\boldsymbol{\sigma} = \frac{1}{\mathcal{V}} \int_{\partial\mathcal{B}} \boldsymbol{t} \otimes \boldsymbol{x} \, \mathrm{d}A \tag{2.9}$$

where $\mathcal{V}$ is the RVE volume, $\boldsymbol{n}$ the outward unit normal on $\partial\mathcal{B}$, and $\boldsymbol{t} = \boldsymbol{\sigma} \cdot \boldsymbol{n}$ the surface traction.

The Hill-Mandel condition then requires:

$$\int_{\partial\mathcal{B}} \left[\delta\boldsymbol{u} - \delta^{\mathrm{M}}\boldsymbol{\varepsilon} \cdot \boldsymbol{x}\right] \cdot \left[\boldsymbol{t} - {}^{\mathrm{M}}\boldsymbol{\sigma} \cdot \boldsymbol{n}\right] \mathrm{d}A = 0 \tag{2.10}$$

This is satisfied by linear displacement (Dirichlet) boundary conditions, uniform traction (Neumann) boundary conditions, or periodic boundary conditions [4].

2.1.5 Eshelby Problem and the Mori-Tanaka Assumption

The Eshelby inclusion problem [8] considers a single ellipsoidal inhomogeneity with stiffness $\mathbb{C}^{(1)}$ embedded in an infinite matrix with stiffness $\mathbb{C}^{(2)}$, subject to a uniform far-field strain $\boldsymbol{\varepsilon}^{(0)}$. Eshelby's fundamental result is that the strain inside the inhomogeneity is uniform, and the strain concentration tensor $\mathbb{T}$ satisfies:

$$\boldsymbol{\varepsilon}^{(1)} = \mathbb{T}^{(1)} : \boldsymbol{\varepsilon}^{(0)} \tag{2.11}$$

The Mori-Tanaka method [9], [10] extends this result to finite inclusion volume fractions by a mean-field assumption: each inclusion experiences the *average matrix strain* $\boldsymbol{\varepsilon}^{(2)}$ (rather than the macroscopic strain) as its far-field condition. The effective stiffness then follows from the concentration tensors and volume fractions.

For composites with imperfect interfaces, the classical Eshelby formula (2.11) no longer applies directly, because the interface conditions modify the stress and displacement fields around the inclusion. In this case, the concentration tensors are computed by solving boundary value problems that incorporate the specific interface constitutive law a procedure detailed in Chapter 3.

2.2 Three-Layered Interphase Model

In many composite systems, the transition from inclusion to matrix is not abrupt but occurs through a graded interphase region with varying material properties. The idea of replacing a multi-coated inclusion by an equivalent homogeneous one dates back to the works of Hervé and Zaoui [20], [21] and Benveniste et al. [19]. The formulations and notation used throughout this section follow the report by Chatzigeorgiou et al. [41]. The three-layered interphase model handles this by discretising the interphase into N coating layers, each with piecewise-constant material properties. An inhomogeneity (core) with stiffness $\mathbb{C}^{(1)}$ is surrounded by coating layers with stiffnesses $\mathbb{C}^{(2)}, \mathbb{C}^{(3)}, \ldots, \mathbb{C}^{(N+1)}$, where N is number of coatings. All of this is embedded in an infinite matrix with stiffness $\mathbb{C}^{(5)}$ subject to a uniform far-field strain. In the present work, we use $N = 3$ coating layers, which convergence studies show is sufficient for most engineering interphase profiles. To avoid working with a large number of material phases, the core and its surrounding coatings are replaced by a single equivalent inhomogeneity with effective stiffness $\mathbb{C}^{\text{eq}}$. We then apply Mori-Tanaka homogenization to obtain the effective properties of the composite system.

2.2.1 Average Fields and Volume Fractions

Our goal of the multi-coated inclusion analysis is to determine concentration tensors that relate the average fields in each phase to the applied far-field strain $\boldsymbol{\varepsilon}^{(0)}$.

Strain-Type Concentration Tensor

The strain-type concentration tensor $\mathbb{T}^{(i)}$ satisfies:

$$\langle \boldsymbol{\varepsilon}^{(i)} \rangle = \mathbb{T}^{(i)} : \boldsymbol{\varepsilon}^{(0)}, \quad i = h, 1, 2, \ldots, N \tag{2.12}$$

Stress-Strain Concentration Tensor

The stress-strain concentration tensor $\mathbb{H}^{(i)}$ satisfies:

$$\langle \boldsymbol{\sigma}^{(i)} \rangle = \mathbb{H}^{(i)} : \boldsymbol{\varepsilon}^{(0)}, \quad i = h, 1, 2, \ldots, N \tag{2.13}$$

For homogenization, we define the average strain and stress in each phase i:

$$\langle \boldsymbol{\varepsilon}^{(i)} \rangle = \frac{1}{V_i} \int_{\Omega_i} \boldsymbol{\varepsilon}(\boldsymbol{x})\, dV, \quad \langle \boldsymbol{\sigma}^{(i)} \rangle = \frac{1}{V_i} \int_{\Omega_i} \boldsymbol{\sigma}(\boldsymbol{x})\, dV \tag{2.14}$$

for $i = 1, 2, 3, 4, 5$.

For phases with constant material properties, these tensors are related by:

$$\mathbb{H}^{(i)} = \mathbb{C}^{(i)} : \mathbb{T}^{(i)} \tag{2.15}$$

Volume Fractions

The volume fractions characterise the relative sizes of the phases within the equivalent inhomogeneity:

$$c_m = \frac{V_m}{\sum_{r=1}^{N+1} V_r}, \quad \phi_m = \frac{\sum_{r=1}^{m} V_r}{\sum_{r=1}^{m+1} V_r} \tag{2.16}$$

where V_m is the volume of phase $m = 1, 2, 3, 4$(core+3 coatings). These can be expressed recursively:

$$\begin{aligned} c_4 &= 1 - \phi_4 \\ c_3 &= \phi_4 - \phi_4\phi_3 \\ c_2 &= \phi_4\phi_3 - \phi_4\phi_3\phi_2 \\ c_1 &= \phi_4\phi_3\phi_2\phi_1 \quad \text{(core)} \end{aligned} \tag{2.17}$$

The geometric formulae differ for the two composite types. For **coated cylindrical fibres** (2D, CCA geometry):

$$c_1 = \frac{r_1^2}{r_{N+1}^2}, \quad c_m = \frac{r_{m+1}^2 - r_m^2}{r_{N+1}^2}, \quad \phi_m = \left(\frac{r_m}{r_{m+1}}\right)^2 \tag{2.18}$$

For **coated spherical particles** (3D, CSA geometry):

$$c_1 = \frac{r_1^3}{r_{N+1}^3}, \quad c_m = \frac{r_{m+1}^3 - r_m^3}{r_{N+1}^3}, \quad \phi_m = \left(\frac{r_m}{r_{m+1}}\right)^3 \tag{2.19}$$

where r_1 is the core radius and r_m denotes the outer radius of coating layer $m = 2, 3, 4$ (inner radius of layer m).

Equivalent Inhomogeneity

For an arbitrary field $\boldsymbol{a}$ (strain or stress type), its value in the equivalent inhomogeneity is:

$$\boldsymbol{a}^{\text{eq}} = \sum_{r=1}^{N+1} c_r \boldsymbol{a}^{(r)} \tag{2.20}$$

where the volume fractions satisfy:

$$\sum_{i=1}^{N+1} c_i = 1 \tag{2.21}$$

The equivalent concentration tensors are then:

$$\mathbb{T}^{\mathrm{eq}} = \sum_{r=1}^{N+1} c_r \mathbb{T}^{(r)} \tag{2.22}$$

$$\mathbb{H}^{\mathrm{eq}} = \sum_{r=1}^{N+1} c_r \mathbb{H}^{(r)} \tag{2.23}$$

These are volume-weighted averages over all sub-phases. The BVPs that determine the individual phase tensors are derived in Chapter 3.

2.2.2 Model Assumptions

The multi-coated Interphase model rests on the following assumptions:

1. **Linear elasticity** in all phases i.e, core, each coating layer, and matrix
2. **Isotropic phase materials**: Each phase is characterised by two Lamé parameters $(\kappa^{(q)}, \mu^{(q)})$, constant within each phase.
3. **Perfect bonding** between adjacent layers which means displacement and traction are continuous across every internal boundary $\partial\Omega_m$.
4. **Single inclusion embedding**: the multicoated inclusion sits in an infinite matrix subject to a uniform far-field strain $\boldsymbol{\varepsilon}^{(0)}$ (Eshelby problem setting).

2.2.3 Coordinate Transformations

The boundary value problems are formulated in cylindrical coordinates (r, θ, z) for fibres and spherical coordinates (r, θ, ϕ) for particles. Since the concentration tensors are expressed in Voigt notation with respect to Cartesian axes, we need transformation matrices that convert between coordinate systems.

Rotation Matrix

A second-order tensor rotator $\boldsymbol{R}$ maps components from the global Cartesian frame (x_1, x_2, x_3) to a local frame:

$$\boldsymbol{R} = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \tag{2.24}$$

For **cylindrical coordinates** (2D, CCA) with the fibre axis along z:

$$\boldsymbol{R}_{\text{cyl}} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2.25}$$

For **spherical coordinates** (3D, CSA):

$$\boldsymbol{R}_{\text{sph}} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \tag{2.26}$$

Voigt Transformation Matrices

The rotator $\boldsymbol{R}$ is represented in Voigt notation by two 6×6 matrices $\boldsymbol{Q}_\varepsilon$ and $\boldsymbol{Q}_\sigma$ for transforming strains and stresses, respectively.

The strain transformation matrix $\boldsymbol{Q}_\varepsilon$ ($\boldsymbol{\varepsilon}' = \boldsymbol{Q}_\varepsilon \cdot \boldsymbol{\varepsilon}$):

$$\boldsymbol{Q}_\varepsilon = \begin{bmatrix} R_{11}^2 & R_{21}^2 & R_{31}^2 & R_{11}R_{21} & R_{11}R_{31} & R_{21}R_{31} \\ R_{12}^2 & R_{22}^2 & R_{32}^2 & R_{12}R_{22} & R_{12}R_{32} & R_{22}R_{32} \\ R_{13}^2 & R_{23}^2 & R_{33}^2 & R_{13}R_{23} & R_{13}R_{33} & R_{23}R_{33} \\ 2R_{11}R_{12} & 2R_{21}R_{22} & 2R_{31}R_{32} & R_{12}R_{21}+R_{11}R_{22} & R_{12}R_{31}+R_{11}R_{32} & R_{22}R_{31}+R_{21}R_{32} \\ 2R_{11}R_{13} & 2R_{21}R_{23} & 2R_{31}R_{33} & R_{13}R_{21}+R_{11}R_{23} & R_{13}R_{31}+R_{11}R_{33} & R_{23}R_{31}+R_{21}R_{33} \\ 2R_{12}R_{13} & 2R_{22}R_{23} & 2R_{32}R_{33} & R_{13}R_{22}+R_{12}R_{23} & R_{13}R_{32}+R_{12}R_{33} & R_{23}R_{32}+R_{22}R_{33} \end{bmatrix} \tag{2.27}$$

The stress transformation matrix $\boldsymbol{Q}_\sigma$ ($\boldsymbol{\sigma}' = \boldsymbol{Q}_\sigma \cdot \boldsymbol{\sigma}$):

$$\boldsymbol{Q}_\sigma = \begin{bmatrix} R_{11}^2 & R_{21}^2 & R_{31}^2 & 2R_{11}R_{21} & 2R_{11}R_{31} & 2R_{21}R_{31} \\ R_{12}^2 & R_{22}^2 & R_{32}^2 & 2R_{12}R_{22} & 2R_{12}R_{32} & 2R_{22}R_{32} \\ R_{13}^2 & R_{23}^2 & R_{33}^2 & 2R_{13}R_{23} & 2R_{13}R_{33} & 2R_{23}R_{33} \\ R_{11}R_{12} & R_{21}R_{22} & R_{31}R_{32} & R_{12}R_{21}+R_{11}R_{22} & R_{12}R_{31}+R_{11}R_{32} & R_{22}R_{31}+R_{21}R_{32} \\ R_{11}R_{13} & R_{21}R_{23} & R_{31}R_{33} & R_{13}R_{21}+R_{11}R_{23} & R_{13}R_{31}+R_{11}R_{33} & R_{23}R_{31}+R_{21}R_{33} \\ R_{12}R_{13} & R_{22}R_{23} & R_{32}R_{33} & R_{13}R_{22}+R_{12}R_{23} & R_{13}R_{32}+R_{12}R_{33} & R_{23}R_{32}+R_{22}R_{33} \end{bmatrix} \tag{2.28}$$

The forward and inverse transformations are:

$$\boldsymbol{\varepsilon}' = \boldsymbol{Q}_\varepsilon \cdot \boldsymbol{\varepsilon}, \quad \boldsymbol{\sigma}' = \boldsymbol{Q}_\sigma \cdot \boldsymbol{\sigma} \tag{2.29}$$

$$\boldsymbol{\varepsilon} = \boldsymbol{Q}_\sigma^T \cdot \boldsymbol{\varepsilon}', \quad \boldsymbol{\sigma} = \boldsymbol{Q}_\varepsilon^T \cdot \boldsymbol{\sigma}' \tag{2.30}$$

A useful identity is $\boldsymbol{Q}_\varepsilon^{-1} = \boldsymbol{Q}_\sigma^T$ and $\boldsymbol{Q}_\sigma^{-1} = \boldsymbol{Q}_\varepsilon^T$, which follows from the invariance of strain energy density $\frac{1}{2}\boldsymbol{\sigma}^T\boldsymbol{\varepsilon}$ under coordinate rotation.

2.2.4 Concentration Tensors

Tensor Structure for Fibre-Reinforced Composites (2D, CCA)

For long cylindrical fibers, we assume the composite exhibits transverse isotropy. The strain concentration tensor in Voigt notation takes the form:

$$\mathbb{T}^{(q)} = \begin{bmatrix} T_\kappa^{(q)} + T_{44}^{(q)} & T_\kappa^{(q)} - T_{44}^{(q)} & T_{13}^{(q)} & 0 & 0 & 0 \\ T_\kappa^{(q)} - T_{44}^{(q)} & T_\kappa^{(q)} + T_{44}^{(q)} & T_{13}^{(q)} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2T_{44}^{(q)} & 0 & 0 \\ 0 & 0 & 0 & 0 & 2T_{55}^{(q)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 2T_{55}^{(q)} \end{bmatrix} \tag{2.31}$$

The upper-left 3×3 block couples in-plane and axial responses, while $T_{44}^{(q)}$ and $T_{55}^{(q)}$ govern transverse and axial shear.
The stress concentration tensor has the corresponding structure:

$$\mathbb{H}^{(q)} = \begin{bmatrix} H_\kappa^{(q)} + H_{44}^{(q)} & H_\kappa^{(q)} - H_{44}^{(q)} & H_{13}^{(q)} & 0 & 0 & 0 \\ H_\kappa^{(q)} - H_{44}^{(q)} & H_\kappa^{(q)} + H_{44}^{(q)} & H_{13}^{(q)} & 0 & 0 & 0 \\ H_{31}^{(q)} & H_{31}^{(q)} & H_{33}^{(q)} & 0 & 0 & 0 \\ 0 & 0 & 0 & H_{44}^{(q)} & 0 & 0 \\ 0 & 0 & 0 & 0 & H_{55}^{(q)} & 0 \\ 0 & 0 & 0 & 0 & 0 & H_{55}^{(q)} \end{bmatrix} \tag{2.32}$$

All the independent components needed for $\mathbb{T}^{(q)}$, and $\mathbb{H}^{(q)}$ are calculated using BVP solutions (Section 3.1.2).

Tensor Structure for Particle-Reinforced Composites (3D, CSA)

For spherical particles, the tensors exhibit full isotropy: The strain concentration tensor in Voigt notation takes the form:

$$\mathbb{T}^{(q)} = \begin{bmatrix} T_\kappa^{(q)} + \frac{4}{3}T_\mu^{(q)} & T_\kappa^{(q)} - \frac{2}{3}T_\mu^{(q)} & T_\kappa^{(q)} - \frac{2}{3}T_\mu^{(q)} & 0 & 0 & 0 \\ T_\kappa^{(q)} - \frac{2}{3}T_\mu^{(q)} & T_\kappa^{(q)} + \frac{4}{3}T_\mu^{(q)} & T_\kappa^{(q)} - \frac{2}{3}T_\mu^{(q)} & 0 & 0 & 0 \\ T_\kappa^{(q)} - \frac{2}{3}T_\mu^{(q)} & T_\kappa^{(q)} - \frac{2}{3}T_\mu^{(q)} & T_\kappa^{(q)} + \frac{4}{3}T_\mu^{(q)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 2T_\mu^{(q)} & 0 & 0 \\ 0 & 0 & 0 & 0 & 2T_\mu^{(q)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 2T_\mu^{(q)} \end{bmatrix} \tag{2.33}$$

The stress concentration tensor has the corresponding structure:

$$\mathbb{H}^{(q)} = \begin{bmatrix} H_\kappa^{(q)} + \frac{4}{3}H_\mu^{(q)} & H_\kappa^{(q)} - \frac{2}{3}H_\mu^{(q)} & H_\kappa^{(q)} - \frac{2}{3}H_\mu^{(q)} & 0 & 0 & 0 \\ H_\kappa^{(q)} - \frac{2}{3}H_\mu^{(q)} & H_\kappa^{(q)} + \frac{4}{3}H_\mu^{(q)} & H_\kappa^{(q)} - \frac{2}{3}H_\mu^{(q)} & 0 & 0 & 0 \\ H_\kappa^{(q)} - \frac{2}{3}H_\mu^{(q)} & H_\kappa^{(q)} - \frac{2}{3}H_\mu^{(q)} & H_\kappa^{(q)} + \frac{4}{3}H_\mu^{(q)} & 0 & 0 & 0 \\ 0 & 0 & 0 & H_\mu^{(q)} & 0 & 0 \\ 0 & 0 & 0 & 0 & H_\mu^{(q)} & 0 \\ 0 & 0 & 0 & 0 & 0 & H_\mu^{(q)} \end{bmatrix} \tag{2.34}$$

All the independent components needed for $\mathbb{T}^{(q)}$, and $\mathbb{H}^{(q)}$ are calculated from the BVP solutions (Section 3.1.3).

2.2.5 Homogenization

With the equivalent concentration tensors in hand, we obtain the stiffness of the multi-coated inhomogeneity:

$$\mathbb{C}^{\mathrm{eq}} = \mathbb{H}^{\mathrm{eq}} : (\mathbb{T}^{\mathrm{eq}})^{-1} \tag{2.35}$$

The tensor $\mathbb{C}^{\mathrm{eq}}$ represents the stiffness of a single homogeneous inclusion that would produce the same average response as the original multi-coated particle. Since $\mathbb{T}^{\mathrm{eq}}$ and $\mathbb{H}^{\mathrm{eq}}$ already encode the full interaction between the applied far-field strain and the layered inclusion including the volume-weighted contributions of all phases which can be substituted directly into the Mori-Tanaka formula. This Mori-Tanaka assumption replaces the far-field strain with the average matrix strain ($\boldsymbol{\varepsilon}^{(0)} \equiv \boldsymbol{\varepsilon}^{(2)}$), and the macroscopic composite stiffness follows as:

$$\boxed{{}^{\mathrm{M}}\mathbb{C} = \Big[(1-f)\mathbb{C}^{(2)} + f\,\mathbb{H}^{\mathrm{eq}}\Big] : \left[(1-f)\mathbb{I} + f\,\mathbb{T}^{\mathrm{eq}}\right]^{-1}} \tag{2.36}$$

where f is the overall volume fraction of inhomogeneities and $\mathbb{C}^{(2)}$ denotes the matrix stiffness. In the Mori-Tanaka formula we follow the standard two-phase convention (inclusion = phase 1, matrix = phase 2); this is distinct from the multi-layered numbering (0 = matrix, 1 = core, $2, \dots, N{+}1$ = coatings) used in the BVP setup above.

2.3 Extended General Interface Model

An alternative to modelling the interphase as a volumetric region is to shrink it to a zero-thickness interface which has its own constitutive laws. This is the idea behind interface models. Instead of discretising the graded region into layers, we can assign material-like parameters directly to the surface separating inclusion and matrix. In this section we develop the Extended General Interface Model (EGIM) [40], which is the most general linear interface model and encompasses all simpler interface models as special cases. The formulations and derivations presented throughout this section follow those of Firooz et al. [40].

2.3.1 Interface Kinematics

Jump and Average Operators

Considering the interface $\mathcal{I}$ splitting microstructures into two disjoint subdomains denoted as $\mathcal{I}^-$ (the minus side of interface) and $\mathcal{I}^+$ (the plus side of interface). For any field ϕ defined on both sides of the interface, we define:

$$[\![\phi]\!] := \phi^+ - \phi^- \quad \text{(jump operator)} \tag{2.37}$$

$$\{\{\phi\}\} := \frac{1}{2}\left(\phi^+ + \phi^-\right) \quad \text{(average operator)} \tag{2.38}$$

Interface Equilibrium

When an interface $\mathcal{I}$ separates two phases, the equilibrium condition on $\mathcal{I}$ reads:

$$\overline{\mathrm{Div}}\bar{\boldsymbol{\sigma}} + [\![\boldsymbol{\sigma}]\!] \cdot \bar{\boldsymbol{n}} = \mathbf{0}, \quad \text{along } \mathcal{I} \tag{2.39}$$

$$\bar{t} = \{\{\sigma\}\}_{1-\alpha} \cdot \bar{n} = [1-\bar{\alpha}]\,[\sigma^+ \cdot \bar{n}] + \bar{\alpha}\,[\sigma^- \cdot \bar{n}] \quad \text{across } \mathcal{I} \tag{2.40}$$

with $\boldsymbol{\sigma}$and $\bar{\boldsymbol{\sigma}}$ being bulk and the interface stress tensors.

2.3.2 Hierarchy of Interface Models

We present the hierarchy from the simplest to the most general, as each model builds on the one before it.

Perfect Interface

The classical *perfect interface* assumes continuity of both displacement and traction:

$$[\![\boldsymbol{u}]\!] = \boldsymbol{0} \quad \text{(displacement jump)} \tag{2.41}$$

$$[\![\boldsymbol{t}]\!] = [\![\boldsymbol{\sigma} \cdot \bar{\boldsymbol{n}}]\!] = \boldsymbol{0} \quad \text{(traction jump)} \tag{2.42}$$

While convenient, this model cannot capture any interphase physics and fails to represent imperfect adhesion or surface effects.

Cohesive (Spring-Layer) Interface

The cohesive interface model allows displacement jumps while maintaining traction continuity. The interface behaves as a distributed spring:

$$[\![\boldsymbol{u}]\!] \neq \boldsymbol{0} \quad \text{(displacement jump)} \tag{2.43}$$

$$[\![\boldsymbol{t}]\!] = \boldsymbol{0} \quad \text{(traction jump)} \tag{2.44}$$

Elastic (Membrane) Interface

The elastic interface model maintains displacement continuity while allowing traction jumps due to surface stresses:

$$[\![\boldsymbol{u}]\!] = \boldsymbol{0} \quad \text{(displacement jump)} \tag{2.45}$$

$$[\![\boldsymbol{t}]\!] \neq 0 \quad \text{(traction jump)} \tag{2.46}$$

General Interface Model (GIM)

Real interphases often exhibit characteristics of both soft and hard behaviour simultaneously. The General Interface Model unifies the cohesive and elastic models by admitting both displacement and traction jumps:

$$[\![\boldsymbol{u}]\!] \neq \boldsymbol{0} \quad \text{(displacement jump)} \tag{2.47}$$

$$[\![\boldsymbol{t}]\!] \neq \boldsymbol{0} \quad \text{(traction jump)} \tag{2.48}$$

Extended General Interface Model (EGIM)

A crucial limitation of all the above models is the implicit assumption that the interface coincides with the geometric mid-plane of the original interphase i.e, $\bar{\alpha} = 0.5$. The EGIM further generalises this by introducing the position parameter $\bar{\alpha} \in [0, 1]$ that accounts for asymmetric interphases.

The **weighted average operator** for any field ϕ is:

$$\{\{\phi\}\}_{\bar{\alpha}} := \bar{\alpha}\phi^+ + (1 - \bar{\alpha})\phi^- \tag{2.49}$$

Similarly, the complementary weighted average is:

$$\{\{\phi\}\}_{1-\bar{\alpha}} := (1 - \bar{\alpha})\phi^+ + \bar{\alpha}\phi^- \tag{2.50}$$

The **interface displacement $\bar{\boldsymbol{u}}$** and **interface traction $\bar{\boldsymbol{t}}$** are:

$$\bar{\boldsymbol{u}} := \{\{\boldsymbol{u}\}\}_{\bar{\alpha}} = \bar{\alpha}\boldsymbol{u}^+ + (1 - \bar{\alpha})\boldsymbol{u}^- \tag{2.51}$$

$$\bar{\boldsymbol{t}} := \{\{\boldsymbol{\sigma}\}\}_{1-\bar{\alpha}} \cdot \bar{\boldsymbol{n}} = (1 - \bar{\alpha})\,\boldsymbol{\sigma}^+ \cdot \bar{\boldsymbol{n}} + \bar{\alpha}\,\boldsymbol{\sigma}^- \cdot \bar{\boldsymbol{n}} \tag{2.52}$$

The position parameter interpolates between different interface configurations: $\bar{\alpha} = 0.5$ gives the standard GIM (interface at mid-plane), $\bar{\alpha} = 0$ places the interface at the inclusion surface, and $\bar{\alpha} = 1$ places it at the matrix boundary.

2.3.3 Interface Constitutive Relations

The EGIM decomposes interface behaviour into tangential (in-plane) and orthogonal (normal) components.

Fibre Composites (2D, CCA)

For fibre-reinforced composites with a cylindrical interface, the constitutional behaviour reads:

$$\begin{aligned}
\text{tangential:} \quad & \begin{bmatrix} \bar{\sigma}_{\theta\theta} \\ \bar{\sigma}_{zz} \\ \bar{\sigma}_{\theta z} \end{bmatrix} = \begin{bmatrix} \bar{\lambda} + 2\bar{\mu}_{\mathrm{tr}} & \bar{\lambda} & 0 \\ \bar{\lambda} & \bar{\lambda} + 2\bar{\mu}_{\mathrm{tr}} & 0 \\ 0 & 0 & \bar{\mu}_{\mathrm{ax}} \end{bmatrix} \begin{bmatrix} \bar{\varepsilon}_{\theta\theta} \\ \bar{\varepsilon}_{zz} \\ 2\bar{\varepsilon}_{\theta z} \end{bmatrix}, \\
\text{orthogonal:} \quad & \begin{bmatrix} \bar{t}_r \\ \bar{t}_\theta \\ \bar{t}_z \end{bmatrix} = \begin{bmatrix} \bar{k}_r [\![u_r]\!] \\ \bar{k}_\theta [\![u_\theta]\!] \\ \bar{k}_z [\![u_z]\!] \end{bmatrix}.
\end{aligned} \tag{2.53}$$

where $\bar{\lambda}$, $\bar{\mu}_{\text{tr}}$, and $\bar{\mu}_{\text{ax}}$ are the interface Lamé parameters. For an isotropic interface, $\bar{\mu}_{\text{tr}} = \bar{\mu}_{\text{ax}} = \bar{\mu}$.

The strain field on a cylindrical interface at radius r_1 are computed as follows:

$$\bar{\varepsilon}_{\theta\theta} = \frac{1}{r_1}\frac{\partial \bar{u}_\theta}{\partial \theta} + \frac{\bar{u}_r}{r_1}, \quad \bar{\varepsilon}_{zz} = \frac{\partial \bar{u}_z}{\partial z}, \quad 2\bar{\varepsilon}_{\theta z} = \frac{1}{r_1}\frac{\partial \bar{u}_z}{\partial \theta} + \frac{\partial \bar{u}_\theta}{\partial z} \tag{2.54}$$

Particle Composites (3D, CSA)

For particle-reinforced composites with a spherical interface, the constitutive behaviour reads:

$$\begin{aligned} \text{tangential:} \quad & \begin{bmatrix} \bar{\sigma}_{\theta\theta} \\ \bar{\sigma}_{\phi\phi} \\ \bar{\sigma}_{\theta\phi} \end{bmatrix} = \begin{bmatrix} \bar{\lambda} + 2\bar{\mu} & \bar{\lambda} & 0 \\ \bar{\lambda} & \bar{\lambda} + 2\bar{\mu} & 0 \\ 0 & 0 & \bar{\mu} \end{bmatrix} \begin{bmatrix} \bar{\varepsilon}_{\theta\theta} \\ \bar{\varepsilon}_{\phi\phi} \\ 2\bar{\varepsilon}_{\theta\phi} \end{bmatrix}, \\ \text{orthogonal:} \quad & \begin{bmatrix} \bar{t}_r \\ \bar{t}_\theta \\ \bar{t}_\phi \end{bmatrix} = \begin{bmatrix} \bar{k}_r [\![u_r]\!] \\ \bar{k}_\theta [\![u_\theta]\!] \\ \bar{k}_\phi [\![u_\phi]\!] \end{bmatrix}. \end{aligned} \tag{2.55}$$

Due to the full isotropy of the spherical geometry, only two interface moduli ($\bar{\lambda}$, $\bar{\mu}$) govern the tangential response which means there is no distinction between in-plane and out-of-plane shear as in the cylindrical case.

For isotropic interfaces which we are assuming in both 2D and 3D, a single parameter $\bar{k}$ is enough:

$$\bar{k}_r = \bar{k}_\theta = \bar{k}_z = \bar{k} \tag{2.56}$$

Complete EGIM Parameter Set

The complete EGIM is characterised by four independent parameters:

1. $\bar{k}$ — interface orthogonal stiffness
2. $\bar{\lambda}$ — interface first Lamé parameter (coupling between tangential strains)
3. $\bar{\mu}$ — interface shear modulus (resistance to tangential deformation)
4. $\bar{\alpha}$ — interface position parameter

2.3.4 Model Assumptions

The EGIM rests on assumptions different from those of the multi-coated model:

1. **Zero-thickness interface**: the finite-thickness interphase is collapsed onto a surface $\mathcal{I}$ with no volume. All interphase effects are encoded in the surface constitutive law.

2. **Linear interface response**: both the tangential and orthogonal components obey linear constitutive relations with constant parameters.

3. **Isotropic interface**: for the standard model, a single set of four parameters $(\bar{k}, \bar{\lambda}, \bar{\mu}, \bar{\alpha})$ governs the entire interface behaviour.

2.3.5 Modified Hill-Mandel Condition

When interfaces with displacement jumps are present, the standard averaging relations from Section 2.1.4 must be modified to account for discontinuities. The macroscopic strain and stress acquire interface contributions:

$$^{\mathrm{M}}\boldsymbol{\varepsilon} = (1-f)\boldsymbol{\varepsilon}^{(2)} + f\boldsymbol{\varepsilon}^{(1)} + \hat{\boldsymbol{\varepsilon}} \tag{2.57}$$

$$^{\mathrm{M}}\boldsymbol{\sigma} = (1-f)\boldsymbol{\sigma}^{(2)} + f\boldsymbol{\sigma}^{(1)} + \hat{\boldsymbol{\sigma}} \tag{2.58}$$

where f is the inclusion volume fraction, $\boldsymbol{\varepsilon}^{(i)}$ and $\boldsymbol{\sigma}^{(i)}$ are phase averages, and the interface contributions are:

$$\hat{\boldsymbol{\varepsilon}} = \frac{1}{2V}\int_{\mathcal{I}} \left([\![\boldsymbol{u}]\!] \otimes \bar{\boldsymbol{n}} + \bar{\boldsymbol{n}} \otimes [\![\boldsymbol{u}]\!] \right) dA \tag{2.59}$$

$$\hat{\boldsymbol{\sigma}} = \frac{1}{V}\int_{\mathcal{I}} \bar{\boldsymbol{\sigma}}\, dA \tag{2.60}$$

The terms $\hat{\boldsymbol{\varepsilon}}$ and $\hat{\boldsymbol{\sigma}}$ vanish for the interphase model (Section 2.2), where all internal boundaries have perfect bonding and the standard volume averaging is recovered.

2.3.6 Concentration Tensors with Interface Effects

In contrast to the interphase model, where $\mathbb{T}^{\mathrm{eq}}$ and $\mathbb{H}^{\mathrm{eq}}$ are volume-weighted averages over multiple sub-phases, the interface model defines these tensors for a *single* equivalent inhomogeneity with modified boundary conditions at the interface. There is only one inclusion and one matrix—no coating layers—but the interface conditions couple the fields across $\mathcal{I}$.

The concentration tensors satisfy the same relations as before:

$$\boldsymbol{\varepsilon}^{(1)} = \mathbb{T}^{(1)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(1)} = \mathbb{H}^{(1)} : \boldsymbol{\varepsilon}^{(0)} \tag{2.61}$$

$$\boldsymbol{\varepsilon}^{(eq)} = \mathbb{T}^{\mathrm{eq}} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(eq)} = \mathbb{H}^{\mathrm{eq}} : \boldsymbol{\varepsilon}^{(0)} \tag{2.62}$$

but now the components of $\mathbb{T}^{\text{eq}}$ and $\mathbb{H}^{\text{eq}}$ depend on the interface parameters $(\bar{k}, \bar{\lambda}, \bar{\mu}, \bar{\alpha})$ in addition to the bulk material properties. They are obtained by solving interface-modified BVPs, as derived in Chapter 3.

Tensor Structure for Fiber Composites (2D, CCA)

For long cylindrical fibres, the composite exhibits transverse isotropy. The strain concentration tensor for the fiber in Voigt notation takes the block-diagonal form:

$$\mathbb{T}^{(1)} = \begin{bmatrix} T_{11}^{(1)} & T_{11}^{(1)} - T_{44}^{(1)} & T_{13}^{(1)} & 0 & 0 & 0 \\ T_{11}^{(1)} - T_{44}^{(1)} & T_{11}^{(1)} & T_{13}^{(1)} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{44}^{(1)} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{55}^{(1)} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{55}^{(1)} \end{bmatrix} \tag{2.63}$$

The stress concentration tensor has the corresponding structure:

$$\mathbb{H}^{(1)} = \begin{bmatrix} H_{11}^{(1)} & H_{11}^{(1)} - 2H_{44}^{(1)} & H_{13}^{(1)} & 0 & 0 & 0 \\ H_{11}^{(1)} - 2H_{44}^{(1)} & H_{11}^{(1)} & H_{13}^{(1)} & 0 & 0 & 0 \\ H_{31}^{(1)} & H_{31}^{(1)} & H_{33}^{(1)} & 0 & 0 & 0 \\ 0 & 0 & 0 & H_{44}^{(1)} & 0 & 0 \\ 0 & 0 & 0 & 0 & H_{55}^{(1)} & 0 \\ 0 & 0 & 0 & 0 & 0 & H_{55}^{(1)} \end{bmatrix} \tag{2.64}$$

For the equivalent inhomogeneity, the strain concentration tensor reads:

$$\mathbb{T}^{(eq)} = \begin{bmatrix} T_{11}^{(eq)} & T_{11}^{(eq)} - T_{44}^{(eq)} & T_{13}^{(eq)} & 0 & 0 & 0 \\ T_{11}^{(eq)} - T_{44}^{(eq)} & T_{11}^{(eq)} & T_{13}^{(eq)} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{44}^{(eq)} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{55}^{(eq)} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{55}^{(eq)} \end{bmatrix} \tag{2.65}$$

The stress concentration tensor for the equivalent inhomogeneity has the corresponding structure:

$$\mathbb{H}^{(eq)} = \begin{bmatrix} H_{11}^{(eq)} & H_{11}^{(eq)} - 2H_{44}^{(eq)} & H_{13}^{(eq)} & 0 & 0 & 0 \\ H_{11}^{(eq)} - 2H_{44}^{(eq)} & H_{11}^{(eq)} & H_{13}^{(eq)} & 0 & 0 & 0 \\ H_{31}^{(eq)} & H_{31}^{(eq)} & H_{33}^{(eq)} & 0 & 0 & 0 \\ 0 & 0 & 0 & H_{44}^{(eq)} & 0 & 0 \\ 0 & 0 & 0 & 0 & H_{55}^{(eq)} & 0 \\ 0 & 0 & 0 & 0 & 0 & H_{55}^{(eq)} \end{bmatrix} \tag{2.66}$$

The unknowns in these tensors depend on interface parameters rather than on multilayer volume averages.

Tensor Structure for Particle Composites (3D, CSA)

For spherical particles, the composite exhibits full isotropy. The strain concentration tensor for the particle in Voigt notation takes the block-diagonal form:

$$\mathbb{T}^{(1)} = \begin{bmatrix} T_{11}^{(1)} & T_{11}^{(1)} - T_{44}^{(1)} & T_{13}^{(1)} & 0 & 0 & 0 \\ T_{11}^{(1)} - T_{44}^{(1)} & T_{11}^{(1)} & T_{13}^{(1)} & 0 & 0 & 0 \\ T_{11}^{(1)} - T_{44}^{(1)} & T_{11}^{(1)} - T_{44}^{(1)} & T_{11}^{(1)} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{44}^{(1)} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{44}^{(1)} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{44}^{(1)} \end{bmatrix} \tag{2.67}$$

The stress concentration tensor for the particle has the corresponding structure:

$$\mathbb{H}^{(1)} = \begin{bmatrix} H_{11}^{(1)} & H_{11}^{(1)} - 2H_{44}^{(1)} & H_{11}^{(1)} - 2H_{44}^{(1)} & 0 & 0 & 0 \\ H_{11}^{(1)} - 2H_{44}^{(1)} & H_{11}^{(1)} & H_{11}^{(1)} - 2H_{44}^{(1)} & 0 & 0 & 0 \\ H_{11}^{(1)} - 2H_{44}^{(1)} & H_{11}^{(1)} - 2H_{44}^{(1)} & H_{11}^{(1)} & 0 & 0 & 0 \\ 0 & 0 & 0 & H_{44}^{(1)} & 0 & 0 \\ 0 & 0 & 0 & 0 & H_{44}^{(1)} & 0 \\ 0 & 0 & 0 & 0 & 0 & H_{44}^{(1)} \end{bmatrix} \tag{2.68}$$

For the equivalent inhomogeneity, the strain concentration tensor reads:

$$\mathbb{T}^{(eq)} = \begin{bmatrix} T_{11}^{(eq)} & T_{11}^{(eq)} - T_{44}^{(eq)} & T_{11}^{(eq)} - T_{44}^{(eq)} & 0 & 0 & 0 \\ T_{11}^{(eq)} - T_{44}^{(eq)} & T_{11}^{(eq)} & T_{11}^{(eq)} - T_{44}^{(eq)} & 0 & 0 & 0 \\ T_{11}^{(eq)} - T_{44}^{(eq)} & T_{11}^{(eq)} - T_{44}^{(eq)} & T_{11}^{(eq)} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{44}^{(eq)} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{44}^{(eq)} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{44}^{(eq)} \end{bmatrix} \tag{2.69}$$

The stress concentration tensor for the equivalent inhomogeneity has the corresponding structure:

$$\mathbb{H}^{(eq)} = \begin{bmatrix} H_{11}^{(eq)} & H_{11}^{(eq)} - 2H_{44}^{(eq)} & H_{11}^{(eq)} - 2H_{44}^{(eq)} & 0 & 0 & 0 \\ H_{11}^{(eq)} - 2H_{44}^{(eq)} & H_{11}^{(eq)} & H_{11}^{(eq)} - 2H_{44}^{(eq)} & 0 & 0 & 0 \\ H_{11}^{(eq)} - 2H_{44}^{(eq)} & H_{11}^{(eq)} - 2H_{44}^{(eq)} & H_{11}^{(eq)} & 0 & 0 & 0 \\ 0 & 0 & 0 & H_{44}^{(eq)} & 0 & 0 \\ 0 & 0 & 0 & 0 & H_{44}^{(eq)} & 0 \\ 0 & 0 & 0 & 0 & 0 & H_{44}^{(eq)} \end{bmatrix} \tag{2.70}$$

Again, these depend on interface parameters rather than on multi-layer volume averages.

2.3.7 Mori-Tanaka Homogenization with Interfaces

The Mori-Tanaka method assumes each inclusion experiences the average matrix strain as its far-field condition: $\boldsymbol{\varepsilon}^{(0)} \equiv \boldsymbol{\varepsilon}^{(2)}$. With this assumption, the macroscopic stiffness tensor is:

$$\boxed{{}^{\mathrm{M}}\mathbb{C} = \left[(1-f)\mathbb{C}^{(2)} + f\mathbb{H}^{\mathrm{eq}}\right] : \left[(1-f)\mathbb{I} + f\mathbb{T}^{\mathrm{eq}}\right]^{-1}} \tag{2.71}$$

where $\mathbb{T}^{\mathrm{eq}}$ and $\mathbb{H}^{\mathrm{eq}}$ incorporate the interface effects. The interface BVPs solve for a single inclusion with modified boundary conditions—there are no internal coating layers. Both the interphase model (Equation (2.36)) and the interface model use the same Mori-Tanaka formula with their respective equivalent concentration tensors substituted directly.

2.4 Neural Networks and Physics-Informed Neural Networks

The parameter identification approach of Chapter 3 relies on a physics-informed neural network. This section introduces the two core concepts required to follow that development: feedforward networks and the physics-informed training paradigm.

Feedforward Neural Networks. A feedforward neural network maps an input vector $\boldsymbol{x}$ to an output vector $\boldsymbol{y}$ through a sequence of layers. Each layer applies a linear map followed by a nonlinear activation function σ:

$$\boldsymbol{y}^{(\ell)} = \sigma\Big(\boldsymbol{W}^{(\ell)}\boldsymbol{y}^{(\ell-1)} + \boldsymbol{b}^{(\ell)}\Big), \quad \ell = 1, \dots, L \tag{2.72}$$

where $\boldsymbol{W}^{(\ell)}$ is a weight matrix, $\boldsymbol{b}^{(\ell)}$ a bias vector, and $\boldsymbol{y}^{(0)} = \boldsymbol{x}$. The network weights $\boldsymbol{\theta} = \{\boldsymbol{W}^{(\ell)}, \boldsymbol{b}^{(\ell)}\}_\ell$ are calibrated by minimising a loss function $\mathcal{L}(\boldsymbol{\theta})$ that measures the error between predictions and target values, using gradient-based optimisation with a learning rate schedule to allow fine convergence in later training epochs. With sufficient width, such networks are universal approximators, capable of representing any continuous function on a compact domain [42], [43]. For a comprehensive treatment of deep learning fundamentals, the reader is referred to Goodfellow et al. [44].

Physics-Informed Neural Networks. Standard neural networks learn entirely from data. When the system obeys known physical laws, these laws can be embedded directly into the training process. Raissi et al. [36] introduced physics-informed neural networks (PINNs), where the loss function includes a term that penalises violations of the governing equations:

$$\mathcal{L} = \mathcal{L}_{\text{data}}(\boldsymbol{\theta}) + w_{\text{phys}}\, \mathcal{L}_{\text{physics}}(\boldsymbol{\theta}) \tag{2.73}$$

Here $\mathcal{L}_{\text{data}}$ measures the fit to observations, $\mathcal{L}_{\text{physics}}$ measures how well the network outputs satisfy the physical equations, and w_{phys} balances the two contributions. The physics loss restricts the solution space to physically admissible configurations and improves generalisation beyond the training data.

PINNs operate in two modes. In the *forward* mode the network approximates the solution field of a differential equation. In the *inverse* mode the unknowns are the parameters of the governing equations themselves: given observed data, the network recovers the unknown parameters and the physics loss verifies that they reproduce the observations. The present work uses the inverse mode exclusively.

A known challenge in PINN training is that the data and physics loss terms can compete during optimisation; adaptive weighting strategies [45] are used to prevent either term from dominating, and are described in full in Chapter 3.

Connection to the Present Work. The inverse PINN developed in this thesis takes the effective properties of a composite with graded interphase as input and predicts the four unknown EGIM interface parameters $\bar{k}$, $\bar{\lambda}$, $\bar{\mu}$, $\bar{\alpha}$. The physics loss evaluates these predicted parameters through the differentiable EGIM forward model from Section 2.3 and penalises discrepancies against the target properties. Unlike PDE-based PINNs, this setup involves no PDE residual evaluation; the "physics" is the algebraic homogenization model itself, which is exact and fully differentiable. The network we are using is more precisely a *physics-constrained* neural network, since the physical laws enter as algebraic constraints in the loss rather than as PDE residuals. Nevertheless, we retain the term PINN throughout this work, following its broader usage in the literature. Chapter 3 develops the specific architecture, output constraints, training data generation, and loss function for both CCA and CSA composites.

Chapter 3

Methodology

Chapter 2 established the theoretical models, the three-layered interphase model and the Extended General Interface Model together with the concentration tensor framework that connects them. It also introduced the neural network concepts that support the inverse methodology. Both models ultimately produce concentration tensors $\mathbb{T}^{\text{eq}}$ and $\mathbb{H}^{\text{eq}}$ that enter the same Mori-Tanaka formula; the difference lies only in how those tensors are obtained.

This chapter develops the boundary value problems that determine the concentration tensor components, for both fibre-reinforced (CCA) and particle-reinforced (CSA) composites. We first solve the multi-coated interphase BVPs (Section 3.1), then the interface-enhanced Mori-Tanaka BVPs (Section 3.2). Section 3.3 presents the specific network architecture based on the concepts introduced in Section 2.4 – that we use to bridge the gap between these two models. Figure 3.1 shows the overall pipeline.

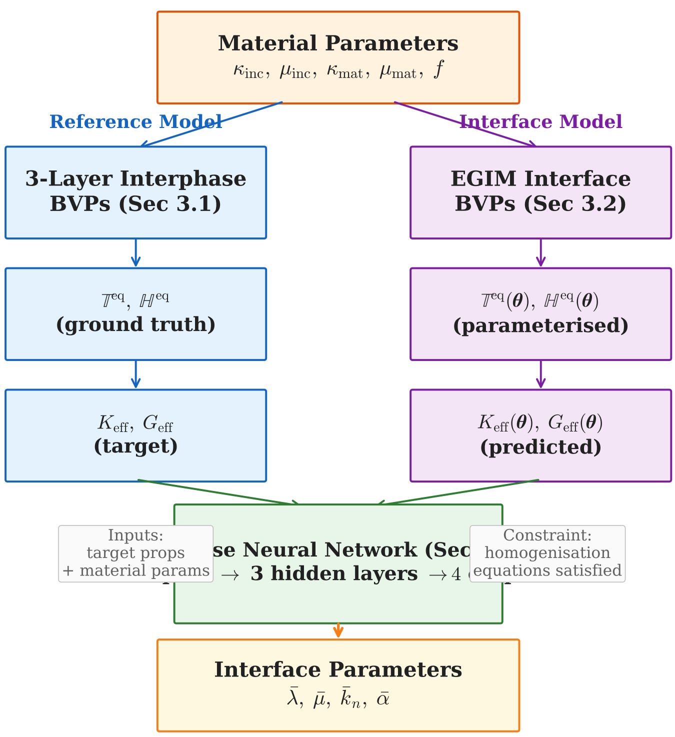

Figure 3.1: Overview of the methodology.

3.1 Three-Layered Interphase Model

The three-layered interphase model introduced in Section 2.2 treats the graded region between inclusion and matrix as 3 coating layers with piecewise-constant properties. To determine the concentration tensors $\mathbb{T}^{(q)}$ and $\mathbb{H}^{(q)}$ for each phase, we solve a set of boundary value problems. Four BVPs are needed for CCA (fibre) composites and two for CSA (particle) composites.

3.1.1 Problem Setup

We consider a single inclusion embedded in an infinite matrix, with the interphase discretised into 3 coating layers between them. For **cylindrical fibres** (CCA), the core/fiber/inclusion has radius r_1 and properties $(\kappa^{(1)}, \mu^{(1)})$. Coating layer m where m = 1,2,3 occupies the annulus $r_m < r < r_{m+1}$ with properties $(\kappa^{(m)}, \mu^{(m)})$, and the matrix has properties $(\kappa^{(5)}, \mu^{(5)})$. The analysis uses cylindrical coordinates (r, θ, z). For **spherical particles** (CSA), the same layered structure applies in spherical coordinates (r, θ, ϕ), with the core particle of radius r_1 coated by 3 concentric shells.

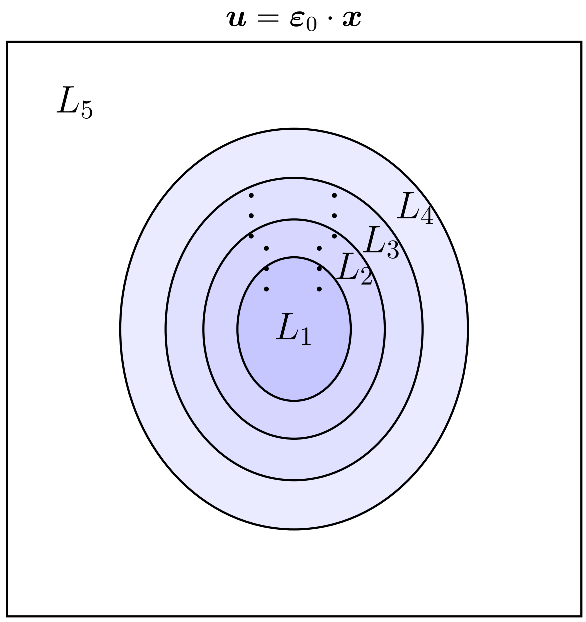

Figure 3.2: Eshelby inhomogeneity problem for a multicoated reinforcement.

To model a graded interphase, each coating layer is assigned elastic properties by interpolating between inclusion and matrix. A weight $w_m \in [0, 1]$ controls how close layer m sits to the inclusion ($w_m = 1$) or the matrix ($w_m = 0$):

$$\kappa^{(m)} = w_m \, \kappa^{(1)} + (1 - w_m) \, \kappa^{(2)}, \quad \mu^{(m)} = w_m \, \mu^{(1)} + (1 - w_m) \, \mu^{(2)}, \quad m = 1, 2, 3 \tag{3.1}$$

A monotonically decreasing sequence such as $(w_1, w_2, w_3) = (0.75,\ 0.50,\ 0.25)$ produces a stiffness profile that grades from the inclusion outward to the matrix. Setting all weights to unity recovers a homogeneous inclusion with no interphase, while setting them to zero collapses the interphase into the matrix; both serve as verification benchmarks in Section 4.1.2. This parameterisation is convenient because a single set of weights generates all coating moduli from the constituent properties, and varying the weights lets us sweep across a wide range of interphase profiles during training data generation.

A uniform far-field strain $\boldsymbol{\varepsilon}^{(0)}$ is applied at infinity (Figure 3.2). In each phase, we write the displacement and stress fields as functions of unknown amplitude coefficients. Continuity of displacement and traction at every internal boundary yields a linear system whose solution gives these amplitudes. Volume averaging over each phase then produces the concentration tensor components. The volume fractions and equivalent tensor formulas from Section 2.2.1 apply directly.

3.1.2 Fibre-Reinforced Composites (CCA)

The transversely isotropic tensor structure (Section 2.2.4) requires four independent BVPs: in-plane bulk, transverse shear, axial shear, and overall axisymmetric loading.

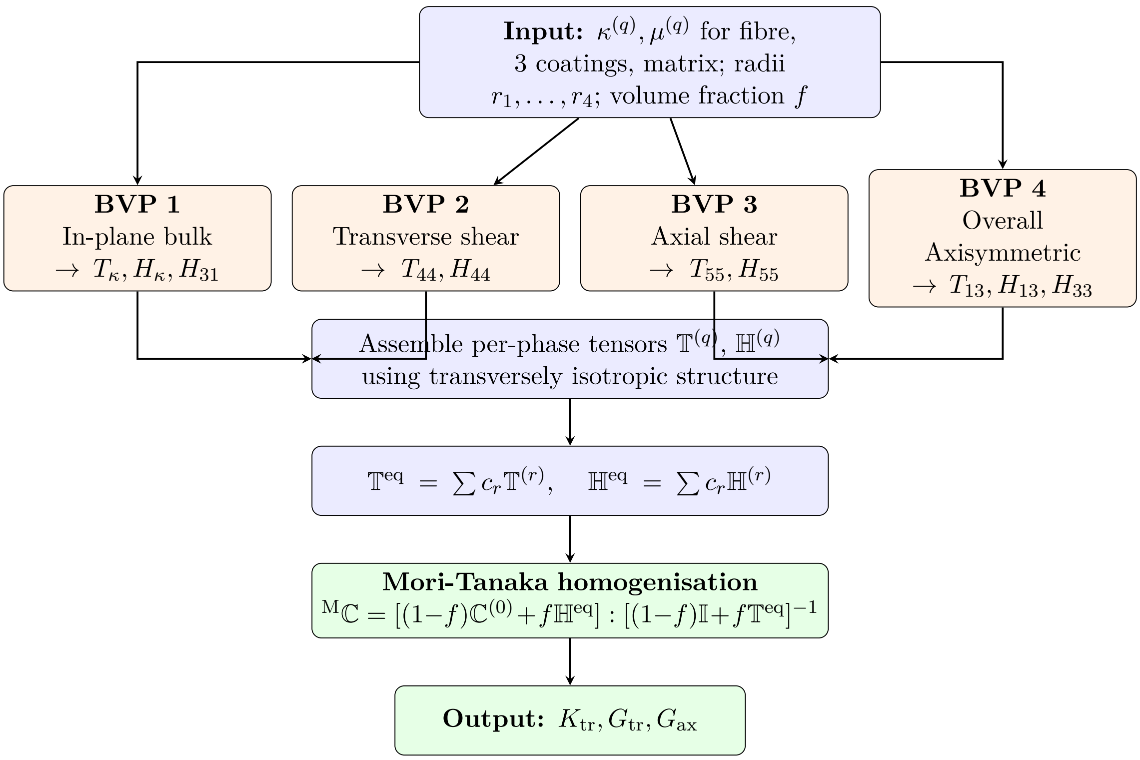

Figure 3.3: Flowchart for fibre-reinforced composites (CCA) using the three-layered interphase model.

BVP 1: In-Plane Bulk Response

Consider RVE subject to far-field displacement $\boldsymbol{u}^{(0)} = [\beta r, 0, 0]^T$ in cylindrical coordinates, producing far-field strain $\boldsymbol{\varepsilon}^{(0)} = [\beta, \beta, 0, 0, 0, 0]^T$, then the displacement field in each phase/constituent is:

$$u_r^{(q)} = \beta \left(X_{2q-1}\, r + \frac{X_{2q}}{r} \right), \quad u_\theta^{(q)} = u_z^{(q)} = 0 \tag{3.2}$$

for all q = 1,2,3,4,5

The radial stress in each phase q is:

$$\sigma_{rr}^{(q)} = 2\beta \left[(\lambda^{(q)} + \mu^{(q)}) X_{2q-1} - \mu^{(q)} \frac{X_{2q}}{r^2} \right] \tag{3.3}$$

with the 10 unknowns (5 layers*2)which can be determined via applying the boundary conditions:

1. Finite displacement at origin: $X_2 = 0$
2. Continuity of displacement at $r = r_1 \implies u_r^{(1)} = u_r^{(2)}$
3. Continuity of traction at $r = r_1 \implies \sigma_{rr}^{(1)} = \sigma_{rr}^{(2)}$
4. Continuity of displacement at $r = r_2 \implies u_r^{(2)} = u_r^{(3)}$
5. Continuity of traction at $r = r_2 \implies \sigma_{rr}^{(2)} = \sigma_{rr}^{(3)}$
6. Continuity of displacement at $r = r_3 \implies u_r^{(3)} = u_r^{(4)}$
7. Continuity of traction at $r = r_3 \implies \sigma_{rr}^{(3)} = \sigma_{rr}^{(4)}$
8. Continuity of displacement at $r = r_4 \implies u_r^{(4)} = u_r^{(5)}$
9. Continuity of traction at $r = r_4 \implies \sigma_{rr}^{(4)} = \sigma_{rr}^{(5)}$
10. Prescribed displacement: $u_r^{(5)}(r \to \infty) = \beta r$ implies $X_9 = 1$

The average strain field over phase q yields:

$$\langle \boldsymbol{\varepsilon}^{(q)} \rangle = \frac{1}{\pi\,(r_b^2 - r_a^2)} \int_0^{2\pi} \int_{r_a}^{r_b} \mathbf{Q}_\varepsilon^T \cdot \varepsilon^{\mathrm{q}}\; dr\, d\theta \tag{3.4}$$

$$= [X_{2q-1}\beta,\, X_{2q-1}\beta,\, 0,\, 0,\, 0,\, 0]^T \tag{3.5}$$

The average stress field over phase q yields:

$$\langle \boldsymbol{\sigma}^{(q)} \rangle = \frac{1}{\pi\,(r_b^2 - r_a^2)} \int_0^{2\pi} \int_{r_a}^{r_b} \mathbf{Q}_\varepsilon^T \cdot \sigma^{\mathrm{q}}\; dr\, d\theta \tag{3.6}$$

$$= [2\kappa^{(q)} X_{2q-1}\beta,\ 2\kappa^{(q)} X_{2q-1}\beta,\ 2(\kappa^{(q)} - \mu^{(q)}) X_{2q-1}\beta,\ 0,\ 0,\ 0]^T \tag{3.7}$$

from Eq 2.61 we find individual interaction tensors

$$\langle \boldsymbol{\varepsilon}^{(q)} \rangle = \mathbb{T}^{(q)} : \boldsymbol{\varepsilon}^{(0)}, \quad \langle \boldsymbol{\sigma}^{(q)} \rangle = \mathbb{H}^{(q)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.8}$$

(Section 2.2.4) The concentration tensor components are:

$$T_\kappa^{(q)} = X_{2q-1}/2 \tag{3.9}$$

$$H_\kappa^{(q)} = \kappa^{(q)} X_{2q-1} \tag{3.10}$$

$$H_{31}^{(q)} = (\kappa^{(q)} - \mu^{(q)}) X_{2q-1} \tag{3.11}$$

where $\kappa^{(q)} = \lambda^{(q)} + \mu^{(q)}$ is the plane-strain bulk modulus.

BVP 2: Transverse Shear Response

For this case far-field displacement $\boldsymbol{u}^{(0)} = [\beta r \sin 2\theta,\ \beta r \cos 2\theta,\ 0]^T$ and the displacement field in each phase is:

$$u_r^{(q)} = \beta r \sin 2\theta \Big(\frac{\kappa^{(q)} - \mu^{(q)}}{2\kappa^{(q)} + \mu^{(q)}} r^2 X_{4(q-1)+4} + X_{4(q-1)+3} - r^4 X_{4(q-1)+1} + \frac{\kappa^{(q)} + \mu^{(q)}}{\mu^{(q)}} r^2 X_{4(q-1)+2}\Big) \tag{3.12}$$

$$u_\theta^{(q)} = \beta r \cos 2\theta (r^2 X_{4(q-1)+4} + X_{4(q-1)+3} + \frac{1}{r^4} X_{4(q-1)+1} + \frac{1}{r^2} X_{4(q-1)+2} \tag{3.13}$$

$$u_z^{(q)} = 0 \tag{3.14}$$

For all q=1,2,3,4,5

$$\sigma_{rr}^{(q)} = \beta \sin 2\theta (2\mu^{(q)} X_{4(q-1)+3} + 6\mu^{(q)} \frac{X_{4(q-1)+1}}{r^4} - 4\kappa^{(q)} \frac{X_{4(q-1)+2}}{r^2}) \tag{3.15}$$

$$\sigma_{r\theta}^{(q)} = \beta \cos 2\theta \Big(\frac{6\kappa^{(q)}\mu^{(q)}}{2\kappa^{(q)} + \mu^{(q)}} r^2 X_{4(q-1)+4} + 2\mu^{(q)} X_{4(q-1)+3} - 6\mu^{(q)} \frac{X_{4(q-1)+1}}{r^4} + 2\kappa^{(q)} \frac{X_{4(q-1)+2}}{r^2}\Big) \tag{3.16}$$

The average strain field over phase q yields:

$$\langle \boldsymbol{\varepsilon}^{(q)} \rangle = [0,\ 0,\ 0,\ \frac{\beta_0\,(3\,X_4\,\kappa\,{r_a}^2 + 3\,X_4\,\kappa\,{r_b}^2 + 4\,X_3\,\kappa + 2\,X_3\,\mu)}{2\,\kappa + \mu},\ 0,\ 0]^T \tag{3.17}$$

The average stress field over phase q yields:

$$\langle \boldsymbol{\sigma}^{(q)} \rangle = \mu^{(q)} \langle \boldsymbol{\varepsilon}^{(q)} \rangle \tag{3.18}$$

Similarly using Eq 2.61 volume-averaged strain in phase q gives:

$$\boxed{\begin{aligned} T_{44}^{(q)} &= \frac{3X_{4(q-1)+4}\kappa^{(q)} r_a^2 + 3X_{4(q-1)+4}\kappa^{(q)} r_b^2 + 4X_{4(q-1)+3}\kappa^{(q)} + 2X_{4(q-1)+3}\mu^{(q)}}{4\cos(2\theta)(2\kappa^{(q)} + \mu^{(q)})} \\ H_{44}^{(q)} &= \mu^{(q)} T_{44}^{(q)} \end{aligned}} \tag{3.19}$$

where the 20 unknowns (5*4)can be determined via applying the boundary conditions:

1. Finite displacement at origin: $X_1 = 0 \quad X_2 = 0$
2. Orthogonal equilibrium at $r = r_1 \implies u_r^{(1)}(r_1) = u_r^{(2)}(r_1), \quad u_\theta^{(1)}(r_1) = u_\theta^{(2)}(r_1)$
3. Tangential equilibrium at $r = r_1 \implies \sigma_{rr}^{(1)}(r_1) = \sigma_{rr}^{(2)}(r_1), \quad \sigma_{r\theta}^{(1)}(r_1) = \sigma_{r\theta}^{(2)}(r_1)$
4. Orthogonal equilibrium at $r = r_2 \implies u_r^{(2)}(r_2) = u_r^{(3)}(r_2), \quad u_\theta^{(2)}(r_2) = u_\theta^{(3)}(r_2)$
5. Tangential equilibrium at $r = r_2 \implies \sigma_{rr}^{(2)}(r_2) = \sigma_{rr}^{(3)}(r_2), \quad \sigma_{r\theta}^{(2)}(r_2) = \sigma_{r\theta}^{(3)}(r_2)$
6. Orthogonal equilibrium at $r = r_3 \implies u_r^{(3)}(r_3) = u_r^{(4)}(r_3), \quad u_\theta^{(3)}(r_3) = u_\theta^{(4)}(r_3)$
7. Tangential equilibrium at $r = r_3 \implies \sigma_{rr}^{(3)}(r_3) = \sigma_{rr}^{(4)}(r_3), \quad \sigma_{r\theta}^{(3)}(r_3) = \sigma_{r\theta}^{(4)}(r_3)$
8. Orthogonal equilibrium at $r = r_4 \implies u_r^{(4)}(r_4) = u_r^{(5)}(r_4), \quad u_\theta^{(4)}(r_4) = u_\theta^{(5)}(r_4)$
9. Tangential equilibrium at $r = r_4 \implies \sigma_{rr}^{(4)}(r_4) = \sigma_{rr}^{(5)}(r_4), \quad \sigma_{r\theta}^{(4)}(r_4) = \sigma_{r\theta}^{(5)}(r_4)$
10. Prescribed displacement: $u_r^{(5)}(r \to \infty) = \beta r$ implies $X_{17} = 1 \quad X_{18} = 1$

BVP 3: Axial Shear Response

For far-field displacement $\boldsymbol{u}^{(0)} = [0, 0, \beta r \cos\theta]^T$, the corresponding displacement field in each phase is:

$$u_z^{(q)}(r,\theta) = \beta r \cos\theta \left(X_{2q-1} + X_{2q}\frac{1}{r^2} \right), \quad u_r^{(q)} = u_\theta^{(q)} = 0 \tag{3.20}$$

The stress vector for each phase q becomes

$$\boldsymbol{\sigma}^{(q)} = [0,\ 0,\ 0,\ 0,\ \beta_0\,\mu\,\cos(\theta)\,\left(X_1 - \frac{X_2}{r^2}\right),\ -\frac{\beta_0\,\mu\,\sin(\theta)\,\left(X_1\,r^2 + X_2\right)}{r^2}]^T \tag{3.21}$$

The average strain field over phase q yields:

$$\langle \boldsymbol{\varepsilon}^{(q)} \rangle = [0,\, 0,\, 0,\, 0, X_{2q-1}\beta,\, 0]^T \tag{3.22}$$

The average stress field over phase q yields:

$$\langle \boldsymbol{\sigma}^{(q)} \rangle = \mu^{(q)} \langle \boldsymbol{\varepsilon}^{(q)} \rangle \tag{3.23}$$

from Eq 2.61 we find individual interaction tensors

$$\langle \boldsymbol{\varepsilon}^{(q)} \rangle = \mathbb{T}^{(q)} : \boldsymbol{\varepsilon}^{(0)}, \quad \langle \boldsymbol{\sigma}^{(q)} \rangle = \mathbb{H}^{(q)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.24}$$

Then the concentration tensor components are:

$$\boxed{\begin{aligned} T_{55}^{(q)} &= X_{2q-1} && (3.25) \\ H_{55}^{(q)} &= \mu^{(q)} X_{2q-1} && (3.26) \end{aligned}}$$

with the 10 unknowns which can be determined via applying the boundary conditions:

1. Finite displacement at origin: $X_2 = 0$
2. Continuity of displacement at $r = r_1 \implies u_z^{(1)} = u_z^{(2)}$
3. Continuity of traction at $r = r_1 \implies \sigma_{rz}^{(1)} = \sigma_{rz}^{(2)}$
4. Continuity of displacement at $r = r_2 \implies u_z^{(2)} = u_z^{(3)}$
5. Continuity of traction at $r = r_2 \implies \sigma_{rr}^{(2)} = \sigma_{rr}^{(3)}$
6. Continuity of displacement at $r = r_3 \implies u_z^{(3)} = u_z^{(4)}$
7. Continuity of traction at $r = r_3 \implies \sigma_{rz}^{(3)} = \sigma_{rz}^{(4)}$
8. Continuity of displacement at $r = r_4 \implies u_z^{(4)} = u_z^{(5)}$
9. Continuity of traction at $r = r_4 \implies \sigma_{rz}^{(4)} = \sigma_{rz}^{(5)}$
10. Prescribed displacement: $u_z^{(5)}(r \to \infty) = \beta r$ implies $X_9 = 1$

BVP 4: Overall Axisymmetric Response

The first three BVPs determine the in-plane concentration tensor components. To complete the transversely isotropic tensor, we need the coupling components $T_{13}^{(q)}$,

$H_{13}^{(q)}$, and $H_{33}^{(q)}$. So for this case, we apply a uniform axial strain $\varepsilon_{zz}^{(0)} = 1$ at the far field, which produces:

$$\boldsymbol{u}^{(0)}(r,\theta,z) = \begin{bmatrix}\beta r\\ 0\\ \beta z\end{bmatrix}, \quad \boldsymbol{\varepsilon}^{(0)} = \begin{bmatrix}\beta & \beta & \beta & 0 & 0 & 0\end{bmatrix}^T \tag{3.27}$$

In each phase, the radial displacement takes the same form as BVP 1 (in-plane bulk),

$$u_r^{(q)} = \beta r(X_{2q-1} + \frac{X_{2q}}{r^2}), \quad u_\theta = 0, \quad u_z = \beta z \tag{3.28}$$

The average strain field over phase q yields:

$$\langle \boldsymbol{\varepsilon}^{(q)} \rangle = [X_{2q-1}\beta,\ X_{2q-1}\beta,\ \beta,\ 0, 0,\ 0]^T \tag{3.29}$$

The average stress field over phase q yields:

$$\langle \boldsymbol{\sigma}^{(q)} \rangle = [\beta\,(\kappa - \mu + 2\,X_1\,\kappa),\ \beta\,(\kappa - \mu + 2\,X_1\,\kappa),\ \beta\,(\kappa + \mu + 2\,X_1\,\kappa - 2\,X_1\,\mu), 0,\ 0,\ 0] \tag{3.30}$$

from Eq 2.61 we find individual interaction tensors

$$\langle \boldsymbol{\varepsilon}^{(q)} \rangle = \mathbb{T}^{(q)} : \boldsymbol{\varepsilon}^{(0)}, \quad \langle \boldsymbol{\sigma}^{(q)} \rangle = \mathbb{H}^{(q)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.31}$$

The linear system for X_{2q-1} and X_{2q} uses the *same* coefficient matrix as BVP 1, but with a different right-hand side arising from the Lamé parameter mismatch between adjacent phases. Then the off-diagonal concentration tensor components are :

$$\boxed{\begin{aligned} T_{13}^{(q)} &= X_{2q-1} - 2T_\kappa & (3.32)\\ H_{13}^{(q)} &= 2\kappa^{(q)}\,X_{2q-1} + \lambda^{(q)} - 2H_\kappa^{(q)} & (3.33)\\ H_{33}^{(q)} &= 2\lambda^{(q)}\,X_{2q-1} + \left(\kappa^{(q)} + \mu^{(q)}\right) - 2H_{31}^{(q)} & (3.34)\end{aligned}}$$

where $T_\kappa^{(q)}$, H_{31} and $H_\kappa^{(q)}$ come from BVPs 1. Now if we substitute all the unknown Tensor components we just found in Eq (2.31) and (2.32) ,we get Individual phase concentration tensors. Once the per-phase tensors are known, the equivalent concentration tensors $\mathbb{T}^{\text{eq}}$ and $\mathbb{H}^{\text{eq}}$ follow from the volume-weighted averages defined in Equations (3.53)–(3.54), using the volume fractions from Equations (2.18) and (2.19).

$$\mathbb{T}^{\mathrm{eq}} = \sum_{r=1}^{N+1} c_r \mathbb{T}^{(r)} \tag{3.35}$$

$$\mathbb{H}^{\mathrm{eq}} = \sum_{r=1}^{N+1} c_r \mathbb{H}^{(r)} \tag{3.36}$$

3.1.3 Particle-Reinforced Composites (CSA)

The isotropic tensor structure (Section 2.2.4) requires two BVPs: hydrostatic and deviatoric loading. The layered geometry and coordinate system are as described in the problem setup above, now in spherical coordinates.

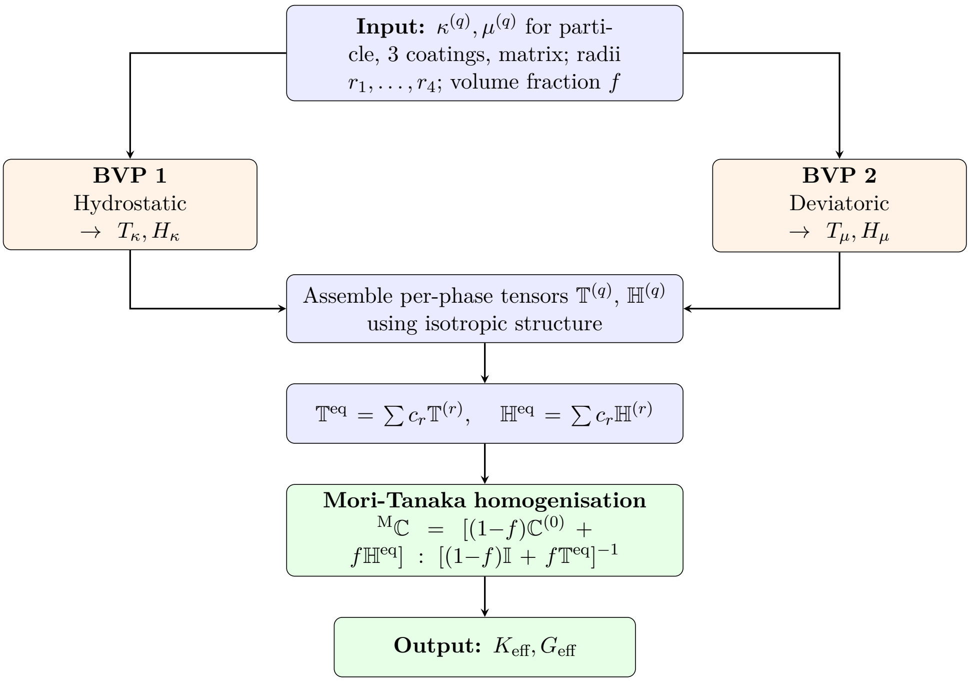

Figure 3.4: Flowchart for particle-reinforced composites (CSA) using the three-layered interphase model.

BVP 1: Hydrostatic Response

Consider the RVE subject to far-field displacement $\boldsymbol{u}^{(0)} = [\beta r, 0, 0]^T$ in spherical coordinates, the displacement field in each phase is:

$$u_r^{(q)} = \beta(X_{2q-1} r + \frac{X_{2q}}{r^2}), \quad u_\theta^{(q)} = u_\phi^{(q)} = 0 \quad q = 1, 2, 3, 4, 5 \tag{3.37}$$

with the 10 unknowns which can be determined from using same boundary conditions as the one we use in In-Plane Response case of our 2D problem. The average strain field vector yields :

$$\langle \varepsilon^{(q)} \rangle = \frac{3}{4\pi\,[r_b^3 - r_a^3]} \int_0^{2\pi} \int_0^{\pi} \int_{r_a}^{r_b} \mathbf{Q}_\varepsilon^T \cdot \varepsilon^{\rm sph}\, r^2 \sin\theta\, dr\, d\theta\, d\phi, \tag{3.38}$$

$$= [X_{2q-1}\beta,\ X_{2q-1}\beta,\ X_{2q-1}\beta,\ 0,\ 0,\ 0]^T \tag{3.39}$$

The Average stress field vector yields

$$\langle \sigma^{(q)} \rangle = \frac{3}{4\pi\,[r_b^3 - r_a^3]} \int_0^{2\pi} \int_0^{\pi} \int_{r_a}^{r_b} \mathbf{Q}_\varepsilon^T \cdot \sigma^{\rm sph}\, r^2 \sin\theta\, dr\, d\theta\, d\phi \tag{3.40}$$

$$= 3\kappa^{(q)} \langle \varepsilon^{(q)} \rangle \tag{3.41}$$

from Eq 2.61 we find individual interaction tensors

$$\langle \boldsymbol{\varepsilon}^{(q)} \rangle = \mathbb{T}^{(q)} : \boldsymbol{\varepsilon}^{(0)}, \quad \langle \boldsymbol{\sigma}^{(q)} \rangle = \mathbb{H}^{(q)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.42}$$

The concentration tensor components are:

$$\boxed{\begin{aligned} T_\kappa^{(q)} &= X_{2q-1} && (3.43) \\ H_\kappa^{(q)} &= 3\kappa^{(q)} X_{2q-1} && (3.44) \end{aligned}}$$

BVP 2: Deviatoric Response

Assume the RVE subject to a deviatoric far-field displacement field as:

$$\boldsymbol{u}^{(0)}(r,\theta,\phi) = \begin{bmatrix} \beta\, r\, \sin^2\theta\, \cos 2\phi \\ \beta\, r\, \sin\theta\, \cos\theta\, \cos 2\phi \\ -\beta\, r\, \sin\theta\, \sin 2\phi \end{bmatrix} \tag{3.45}$$

In each phase, the displacement fields involve four radial modes with exponents ξ_i:

$$u_r^{(q)} = \beta\, r \sum_{i=1}^{4} A_i^{(q)} X_{4(q-1)+i} r^{\xi_i - 1} \sin^2\theta\, \cos 2\phi \tag{3.46}$$

$$u_\theta^{(q)} = \beta\, r \sum_{i=1}^{4} X_{4(q-1)+i} r^{\xi_i - 1} \sin\theta\, \cos\theta\, \cos 2\phi \tag{3.47}$$

$$u_\phi^{(q)} = -\beta\, r \sum_{i=1}^{4} X_{4(q-1)+i} r^{\xi_i - 1} \sin\theta\, \sin 2\phi \tag{3.48}$$

where $\xi_1 = 1$, $\xi_2 = 3$, $\xi_3 = -4$, $\xi_4 = -2$. The amplitude ratio $A_i^{(q)}$ depend on the material properties of phase q:

$$A_1^{(q)} = 1, \quad A_2^{(q)} = \frac{9\kappa^{(q)} - 6\mu^{(q)}}{15\kappa^{(q)} + 11\mu^{(q)}}, \quad A_3^{(q)} = -\frac{3}{2}, \quad A_4^{(q)} = \frac{3(\kappa^{(q)} + \mu^{(q)})}{2\mu^{(q)}} \tag{3.49}$$

where κ and μ here refer to the 3D bulk and shear moduli of the respective phase.

Now for the unknowns that arise from all the layers equals 20 which can be found from the similar Boundary conditions as we used in Transverse Shear response. For the core, $X_3 = X_4 = 0$ to ensure bounded displacement at the origin. The far-field condition requires $X_{17} = 1$ and $X_{18} = 1$. At each interphase $r = r_m$, the continuity of both displacement components and both traction components produces four conditions per interface, yielding the same 20×20 system structure as the CCA transverse shear BVP, but with spherical amplitude ratios. from Eq 2.61 we find individual interaction tensors

$$\langle \boldsymbol{\varepsilon}^{(q)} \rangle = \mathbb{T}^{(q)} : \boldsymbol{\varepsilon}^{(0)}, \quad \langle \boldsymbol{\sigma}^{(q)} \rangle = \mathbb{H}^{(q)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.50}$$

the shear concentration tensor components are:

$$\boxed{\begin{aligned} T_\mu^{(q)} &= X_{4(q-1)+1} + \frac{63\kappa^{(q)} + 21\mu^{(q)}}{5(15\kappa^{(q)} + 11\mu^{(q)})} \cdot \frac{r_b^5 - r_a^5}{r_b^3 - r_a^3} \cdot X_{4(q-1)+2} && (3.51) \\ H_\mu^{(q)} &= 2\mu^{(q)} T_\mu^{(q)} && (3.52) \end{aligned}}$$

The first term is the contribution of the uniform mode ($\xi_1 = 1$), while the second involves the $\xi_2 = 3$ mode weighted by the radii ratio r_b^5/r_a^5. The $\xi_3 = -4$ and $\xi_4 = -2$ modes vanish upon volume averaging Now if we substitute all the unknown Tensor components we just found in Eq (2.33) and (2.34) ,we get Individual phase concentration tensors. Once the per-phase tensors are known, the equivalent concentration tensors $\mathbb{T}^{\text{eq}}$ and $\mathbb{H}^{\text{eq}}$ follow from the volume-weighted averages defined in Equations (3.53)–(3.54), using the volume fractions from Equations (2.18) and (2.19).

$$\mathbb{T}^{\text{eq}} = \sum_{r=1}^{N+1} c_r \mathbb{T}^{(r)} \tag{3.53}$$

$$\mathbb{H}^{\text{eq}} = \sum_{r=1}^{N+1} c_r \mathbb{H}^{(r)} \tag{3.54}$$

3.2 Extended General Interface Model with Mori-Tanaka

Where the interphase model (Section 3.1) uses multiple coating layers, the EGIM collapses the graded region onto a single zero-thickness surface at $r = r_1$, governed by four parameters $(\bar{k}, \bar{\lambda}, \bar{\mu}, \bar{\alpha})$ defined in Section 2.3. The displacement ansätz from Section 3.1 carry over directly, but the perfect-bonding conditions at $r = r_1$ are replaced by the EGIM cohesive and elastic jump conditions from Section 2.3.3.

3.2.1 Problem Setup

We consider a single inclusion (fibre or particle) with properties $(\kappa^{(1)}, \mu^{(1)})$ embedded directly in the matrix $(\kappa^{(2)}, \mu^{(2)})$ meaning there are no coating layers. This applies for both cylindrical fibers with coordinates(r, θ, z) and for spherical particles (CSA) with coordinates (r, θ, ϕ), A uniform far-field strain $\varepsilon^{(0)}$ is applied at infinity. At the interface $r = r_1$, unlike the 3 coated Interphase model, the interface constitutive laws applies which were defined in Section 2.3.3. Solving the resulting BVPs yields the equivalent concentration tensors $\mathbb{T}^{\text{eq}}$ and $\mathbb{H}^{\text{eq}}$ as functions of $(\bar{k}, \bar{\lambda}, \bar{\mu}, \bar{\alpha})$ and the constituent properties.

3.2.2 Fibre-Reinforced Composites (CCA)

For cylindrical fibres, the transversely isotropic concentration tensor (Section 2.2.4) requires four independent BVPs: in-plane bulk, transverse shear, axial shear, and overall axisymmetric loading. Each BVP uses the same displacement field as the corresponding interphase BVP in Section 3.1.2, but with only two phases ($q = 1$ for inclusion, $q = 2$ for matrix).

BVP 1: In-Plane Bulk Response

Consider an RVE subject to a far field displacement $\boldsymbol{u}^{(0)} = [\beta r, 0, 0]^T$, then the displacement field takes the form:

$$u_r^{(q)} = \beta r \left(X_{2q-1} + \frac{X_{2q}}{(r/r_1)^2} \right), \quad u_\theta^{(q)} = u_z^{(q)} = 0 \quad \text{for q = 1,2} \tag{3.55}$$

resulting in 4 unknowns that can be determined by imposing boundary conditions along with interface conditions as follows,

- finite displacement at $r = 0$:

$$u_r^{(1)}(r = 0) \not\to \infty \;\Rightarrow\; X_2 = 0, \tag{47}$$

- orthogonal equilibrium at $r = r_1$:

$$\bar{t}_r = \bar{k}_r [\![u_r]\!] \;\Rightarrow\; [1 - \bar{\alpha}]\, \sigma_{rr}^{(2)}(r_1) + \bar{\alpha}\, \sigma_{rr}^{(1)}(r_1) = \bar{k}_r \left[u_r^{(2)}(r_1) - u_r^{(1)}(r_1)\right], \tag{48}$$

- tangential equilibrium at $r = r_1$:

$$\left[\overline{\operatorname{div}\bar{\sigma}}\right]_r + [\![t_r]\!] = 0 \;\Rightarrow\; \frac{1}{r_1}\bar{\sigma}_{\theta\theta} + \sigma_{rr}^{(2)}(r_1) - \sigma_{rr}^{(1)}(r_1) = 0, \tag{49}$$

- prescribed displacement at $r \to \infty$:

$$u_r^{(2)}(r \to \infty) = \beta r \;\Rightarrow\; X_3 = 1. \tag{50}$$

The two conditions in Eqs 48 and 49 yield a 2×2 linear system for the unknowns $\{X_1, X_4\}$; see [40] for the explicit coefficient matrix. from Eq 2.61 we find individual interaction tensors

$$\boldsymbol{\varepsilon}^{(1)} = \mathbb{T}^{(1)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(1)} = \mathbb{H}^{(1)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.56}$$

$$\boldsymbol{\varepsilon}^{(eq)} = \mathbb{T}^{(eq)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(eq)} = \mathbb{H}^{(eq)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.57}$$

These equations leads to following concentration tensor components:

$$\boxed{\begin{aligned} T_{11}^{1} &= \frac{T_{44}^{(1)} + X_1}{2} && (3.58)\\ T_{11}^{\mathrm{eq}} &= \frac{T_{44}^{(eq)} + X_4}{2} && (3.59)\\ H_{11}^{\mathrm{eq}} &= \kappa^{(2)} - \mu^{(2)} X_4 + H_{44}^{(eq)} && (3.60)\\ H_{31}^{\mathrm{eq}} &= \lambda^{(1)} X_1 && (3.61) \end{aligned}}$$

where $\kappa^{(q)} = \lambda^{(q)} + \mu^{(q)}$ is the plane-strain bulk modulus and the unknowns X_1 and X_4 are the solutions of solving Eqs 48 and 49.

BVP 2: Transverse Shear Response

For this case, the far field displacement / strain applied to the RVE read $\boldsymbol{u}^{(0)} = [\beta r \sin 2\theta,\ \beta r \cos 2\theta,\ 0]^T$. Then the displacement field takes the same form as Equations (3.12)–(3.14), but for only two layers so we have eight unknowns $X_1, \ldots, X_8$ (four per phase) to solve. With $X_3 = X_4 = 0$ (finite at the origin) and far-field conditions on the matrix coefficients says $X_5 = 1$ and $X_6 = 1$, then the 4 unknowns can be found by solving the EGIM conditions at $r = r_1$ are:

$$\text{Orthogonal Equilibrium in r dir:} \quad \bar{\boldsymbol{t}_r} = \bar{k_r}[\![u_r(r_1)]\!] \tag{3.62}$$

$$\text{Orthogonal Equilibrium in } \theta \text{ dir:} \quad \bar{\boldsymbol{t}_\theta} = \bar{k_\theta}[\![u_\theta(r_1)]\!] \tag{3.63}$$

$$\text{Tangential Equilibrium in r dir:} \quad -\frac{\bar{\sigma}_{\theta\theta}}{r_1} + [\![\sigma_{rr}(r_1)]\!] = 0 \tag{3.64}$$

$$\text{Tangential Equilibrium in } \theta \text{ dir:} \quad \frac{1}{r_1}\frac{\partial\bar{\sigma}_{\theta\theta}}{\partial\theta} + [\![\sigma_{r\theta}(r_1)]\!] = 0 \tag{3.65}$$

These four conditions, together with the boundary conditions, yield a 4×4 linear system for the free unknowns; see [40] for the explicit system. from Eq 2.61 we find individual interaction tensors

$$\boldsymbol{\varepsilon}^{(1)} = \mathbb{T}^{(1)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(1)} = \mathbb{H}^{(1)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.66}$$

$$\boldsymbol{\varepsilon}^{(eq)} = \mathbb{T}^{(eq)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(eq)} = \mathbb{H}^{(eq)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.67}$$

These equations leads to following concentration tensor components:

$$\boxed{\begin{aligned} T_{44}^{(1)} &= \frac{3\kappa^{(1)}}{4\kappa^{(1)} + 2\mu_{\text{tr}}^{(1)}} X_1 + X_2, && (3.68) \\ T_{44}^{\text{eq}} &= 1 + \frac{\kappa^{(2)} + 2\mu^{(2)}}{2\mu^{(2)}} X_8 && (3.69) \\ H_{44}^{\text{eq}} &= \mu^{(2)} - \frac{\kappa^{(2)}}{2} X_8 && (3.70) \end{aligned}}$$

BVP 3: Axial Shear Response

For this case, we consider an RVE subject to a far field displacement $\boldsymbol{u}^{(0)} = [0, 0, \beta r\cos\theta]^T$, then the displacement field takes the form :

$$u_z^{(q)} = \beta r\cos\theta\left(X_{2q-1} + \frac{X_{2q}}{(r/r_1)^2}\right), \quad u_\theta^{(q)} = u_r^{(q)} = 0 \quad \text{for q = 1,2} \tag{3.71}$$

resulting in 4 unknowns X_1, X_2, X_3, X_4 that can be determined via imposing finite displacement at origin($X_2 = 0$) and prescribed displacement at $r \to \infty(X_3 =$

1), along with the following interface conditions at $r = r_1$:

$$\text{orthogonal equilibrium:} \quad \bar{t}_z = \bar{k}[\![u_z]\!] \tag{3.72}$$

$$\text{Tangential equilinrium:} \quad \left[\overline{\text{div}\,\bar{\sigma}}\right]_z + [\![t_z]\!] = 0 \tag{3.73}$$

from Eq 2.61 we have

$$\boldsymbol{\varepsilon}^{(1)} = \mathbb{T}^{(1)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(1)} = \mathbb{H}^{(1)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.74}$$

$$\boldsymbol{\varepsilon}^{(eq)} = \mathbb{T}^{(eq)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(eq)} = \mathbb{H}^{(eq)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.75}$$

These equations leads to following concentration tensor components:

$$\boxed{\begin{aligned} T_{55}^{1} &= X_1 && (3.76) \\ T_{55}^{\text{eq}} &= 1 + X_4 && (3.77) \\ H_{55}^{\text{eq}} &= \mu_{ax}^{(2)}(1 - X_4) && (3.78) \end{aligned}}$$

The unknowns X_1 and X_4 are solution of 2×2 system we solved using the equations 3.71 and 3.72

BVP 4: Overall Axisymmetric Response

for this case, we consider an RVE subject to a far field displacement $\boldsymbol{u}^{(0)} = [\beta r, 0, \beta z]^T$, then the displacement field takes the same form as Equation (3.79).

$$u_r^{(q)} = \beta r \left(X_{2q-1} + \frac{X_{2q}}{(r/r_1)^2} \right), \quad u_\theta^{(q)} = 0 \quad u_z^{(q)} = \beta z \quad \text{for q = 1,2} \tag{3.79}$$

since u_z is continuous and does not contribute to the radial interface conditions, the unknowns X_1 and X_4 are the same as in BVP 1 . from Eq 2.61

$$\boldsymbol{\varepsilon}^{(1)} = \mathbb{T}^{(1)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(1)} = \mathbb{H}^{(1)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.80}$$

$$\boldsymbol{\varepsilon}^{(eq)} = \mathbb{T}^{(eq)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(eq)} = \mathbb{H}^{(eq)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.81}$$

These equations leads to following concentration tensor components:

$$\boxed{\begin{aligned} T_{13}^{(1)} &= X_4 + T_{44}^{(1)} - 2T_{11}^{(1)} \quad &(3.82)\\ T_{13}^{(\mathrm{eq})} &= X_4 + T_{44}^{(\mathrm{eq})} - 2T_{11}^{(1)} \quad &(3.83)\\ H_{13}^{(\mathrm{eq})} &= 2\kappa^{(2)} - 2\mu_{tr}^{(2)} X_4 + \lambda^{(2)} + 2H_{44}^{(eq)} - 2H_{11}^{(eq)} \quad &(3.84)\\ H_{33}^{(\mathrm{eq})} &= \kappa^{(1)} + \mu_{tr}^{(1)} + 2\lambda^{(1)} X_1 + \frac{4\bar{\mu}_{tr}}{r_1} - 2H_{31}^{(\mathrm{eq})} \quad &(3.85) \end{aligned}}$$

where X_1, X_4 are from BVP 1, T_{44}^{eq} is from BVP 2, H_{31}^{eq} is from BVP 1, and $T_{11}^{(1)} = \frac{1}{2}(X_1 + T_{44}^{(1)})$ combines the inclusion unknowns from BVPs 1 and 2 (see Equation (**??**) below for the relation between T_{11} and T_κ, T_{44}). Now that we have all unknown coefficients of $T^{(1)}$, $H^{(1)}$, $T^{(eq)}$ and $H^{(eq)}$, we use them to form the Strain and Stress Interaction Tensors of Equivalent inhomogeneity which will later be used in Homogenization.

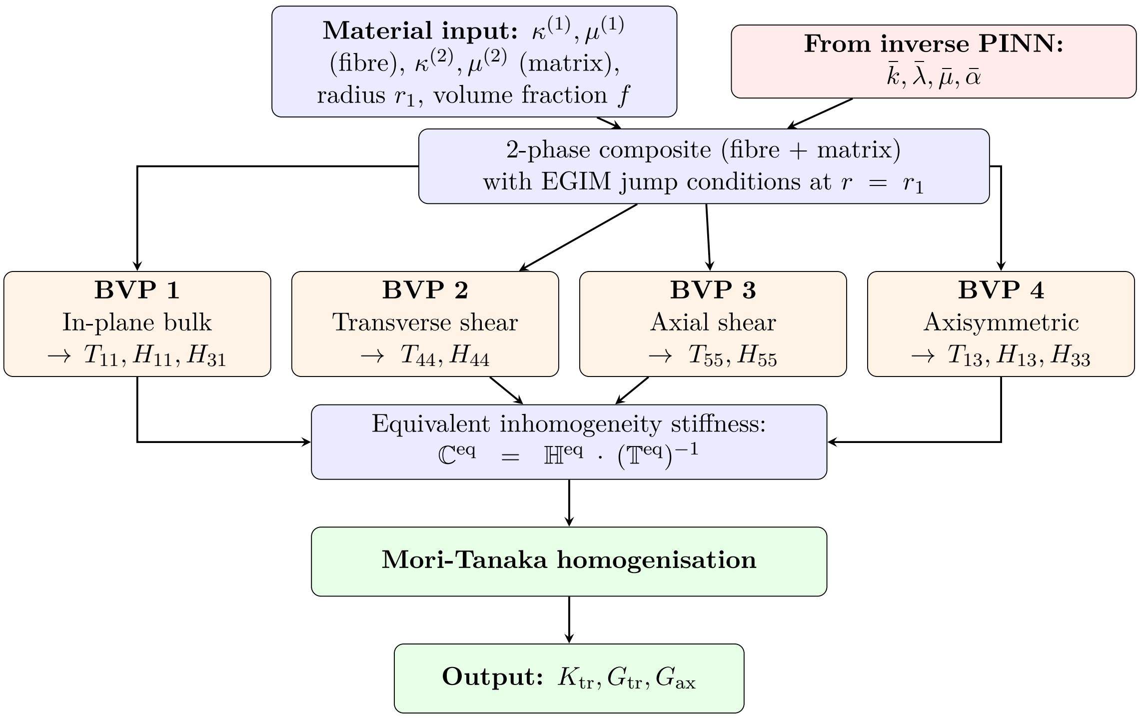

Figure 3.5: Flowchart for fibre-reinforced composites (CCA) using the Extended General Interface Model.

3.2.3 Particle-Reinforced Composites (CSA)

For spherical particles, the isotropic concentration tensor (Section 2.2.4) requires two BVPs: hydrostatic and deviatoric. Each BVP uses the same displacement field as the corresponding interphase BVP in Section 3.1.3, but with only two phases

($q = 1$ for inclusion, $q = 2$ for matrix).

BVP 1: Hydrostatic Response

Consider the RVE subject to far-field displacement $\boldsymbol{u}^{(0)} = [\beta r, 0, 0]^T$ in spherical coordinates, the displacement field in each phase is:

$$u_r^{(q)} = \beta r(X_{2q-1} + \frac{X_{2q}}{(r/r_1)^3}), \quad u_\theta^{(q)} = u_\phi^{(q)} = 0 \quad q = 1, 2 \tag{3.86}$$

with the 4 unknowns which can be determined from using same boundary conditions as the one we use in In-Plane Response case of our 2D EGIM model. from Eq 2.61

$$\boldsymbol{\varepsilon}^{(1)} = \mathbb{T}^{(1)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(1)} = \mathbb{H}^{(1)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.87}$$

$$\boldsymbol{\varepsilon}^{(eq)} = \mathbb{T}^{(eq)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(eq)} = \mathbb{H}^{(eq)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.88}$$

which yields

$$\boxed{\begin{aligned} 3T_{11}^{(1)} - 2T_{44}^{(1)} &= X_1, && (3.89)\\ 3T_{11}^{(\text{eq})} - 2T_{44}^{(\text{eq})} &= [1 + X_4], && (3.90)\\ 3H_{11}^{(\text{eq})} - 4H_{44}^{(\text{eq})} &= [3\kappa^{(2)} - 4\mu^{(2)}]\, X_4. && (3.91) \end{aligned}}$$

where $\kappa^{(q)} = \lambda^{(q)} + \frac{2}{3}\mu^{(q)}$ is the three-dimensional bulk modulus and the Unknowns X_1 and X_4 are solved using the boundary conditions.

BVP 2: Deviatoric Response

Assume the RVE is subject to a deviatoric far field displacement

$$\boldsymbol{u}^{(0)}(r, \theta, \phi) = \begin{bmatrix} \beta\, r\, \sin^2\theta\, \cos 2\phi \\ \beta\, r\, \sin\theta\, \cos\theta\, \cos 2\phi \\ -\beta\, r\, \sin\theta\, \sin 2\phi \end{bmatrix} \tag{3.92}$$

The displacement field takes the same form as Equations (3.46)–(3.47), with eight unknowns. With finite displacement at origin($X_3 = X_4 = 0$) and far-field conditions on the matrix ($X_5 = 1$ and $X_6 = 1$), we apply the same EGIM conditions as we did in Transverse shear response to solve for the 4 unknowns left

from Eq 2.61

$$\boldsymbol{\varepsilon}^{(1)} = \mathbb{T}^{(1)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(1)} = \mathbb{H}^{(1)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.93}$$

$$\boldsymbol{\varepsilon}^{(eq)} = \mathbb{T}^{(eq)} : \boldsymbol{\varepsilon}^{(0)}, \quad \boldsymbol{\sigma}^{(eq)} = \mathbb{H}^{(eq)} : \boldsymbol{\varepsilon}^{(0)} \tag{3.94}$$

which yields

$$\boxed{\begin{aligned} 2T_{44}^{(1)} &= \frac{1}{5}\left[5X_1 - 7\left(1 + 3\frac{\kappa^{(1)}}{\mu^{(1)}}\right)X_2\right], && (3.95)\\ 2T_{44}^{(\mathrm{eq})} &= \frac{1}{5}\left[5 + 6\left(2 + \frac{\kappa^{(2)}}{\mu^{(2)}}\right)X_8\right], && (3.96)\\ 2H_{44}^{(\mathrm{eq})} &= \frac{1}{5}\left[10\mu^{(2)} - 2\left(9\kappa^{(2)} + 8\mu^{(2)}\right)X_8\right] && (3.97)\end{aligned}}$$

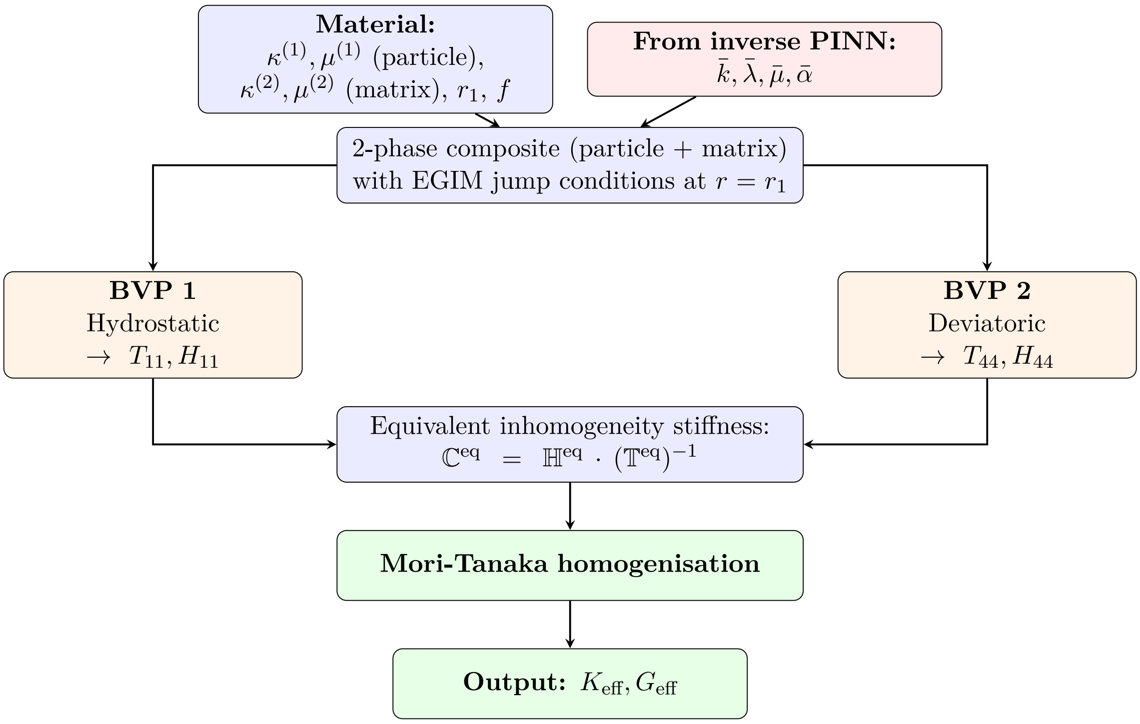

Figure 3.6: Flowchart for particle-reinforced composites (CSA) using the Extended General Interface Model.

3.2.4 Tensor Assembly and Effective Properties

The BVP solutions give the full concentration tensors whose Voigt structure was defined in Section 2.2.4 (CCA) and Section 2.2.4 (CSA). The equivalent stiffness of the equivalent inhomogeneity is:

$$\boxed{\mathbb{C}^{\mathrm{eq}} = \mathbb{H}^{\mathrm{eq}} \cdot (\mathbb{T}^{\mathrm{eq}})^{-1}} \tag{3.98}$$

This $\mathbb{C}^{\mathrm{eq}}$ represents the stiffness of a homogeneous inclusion that produces the same far-field perturbation as the original inclusion surrounded by the imperfect interface. The macroscopic effective stiffness is then obtained from the Mori-Tanaka formula

(Equation (2.71)) using $\mathbb{T}^{\text{eq}}$ and $\mathbb{H}^{\text{eq}}$:

$$^{\text{M}}\mathbb{C} = \left[(1-f)\mathbb{C}^{(2)} + f\mathbb{H}^{\text{eq}}\right] : \left[(1-f)\mathbb{I} + f\mathbb{T}^{\text{eq}}\right]^{-1} \tag{3.99}$$

3.3 Neural Network Architectures

This section describes the neural network architecture used for the main problem of the present work: given the effective properties produced by the three-layered interphase model, find the EGIM interface parameters that reproduce them. We train an inverse network that takes effective moduli and material properties as input and predicts the four unknown interface parameters $\bar{k}$, $\bar{\lambda}$, $\bar{\mu}$, and $\bar{\alpha}$. A physics-informed loss ensures that the predicted parameters, when passed through the EGIM model, reconstruct the target properties.

3.3.1 Inverse Parameter Identification Network

The inverse network addresses the core question: what EGIM parameters produce effective properties matching those of the three-layered interphase model? For a given composite configuration, the interphase BVP solver yields target properties (e.g. K_{tr}, G_{tr}, G_{ax} for CCA). The network must find interface parameters that, when substituted into the EGIM forward model, reproduce these targets as closely as possible.

Input–output specification. For CCA composites, the network takes 8 inputs: the three target effective moduli $(K_{\text{tr}}, G_{\text{tr}}, G_{\text{ax}})$ together with the elastic moduli of inclusion and matrix $(\kappa^{(1)}, \mu^{(1)}, \kappa^{(2)}, \mu^{(2)})$ and the volume fraction f. The network outputs four EGIM parameters: normal stiffness $\bar{k}$, surface Lamé parameter $\bar{\lambda}$, surface shear modulus $\bar{\mu}$, and the distance parameter $\bar{\alpha}$.

For CSA composites, the input reduces to 7 features (two isotropic target moduli K_{eff}, G_{eff} plus material properties and volume fraction), while the output dimension remains four.

Architecture. The network has three fully-connected hidden layers with 256 neurons each, tanh activations, and Xavier normal initialisation [46]. Physical constraints are enforced directly on the outputs: softplus for $\bar{\lambda}$ and $\bar{\mu}$ (non-negativity of surface elastic moduli), an exponential mapping clamped to $[-2, 8.5]$ for $\bar{k}$ (covering $[1, 5000]$, appropriate for a stiffness parameter spanning several orders of magnitude), and a sigmoid for $\bar{\alpha} \in [0, 1]$. The total parameter count is approximately 135 000. Tanh activation is chosen because the output parameters are bounded and continuous: it is smooth everywhere and zero-centred, which suits both gradient flow and the bounded target space. Three hidden layers of 256 neurons provide sufficient capacity for the $8 \to 4$ mapping. Figure 3.7 illustrates the complete architecture.

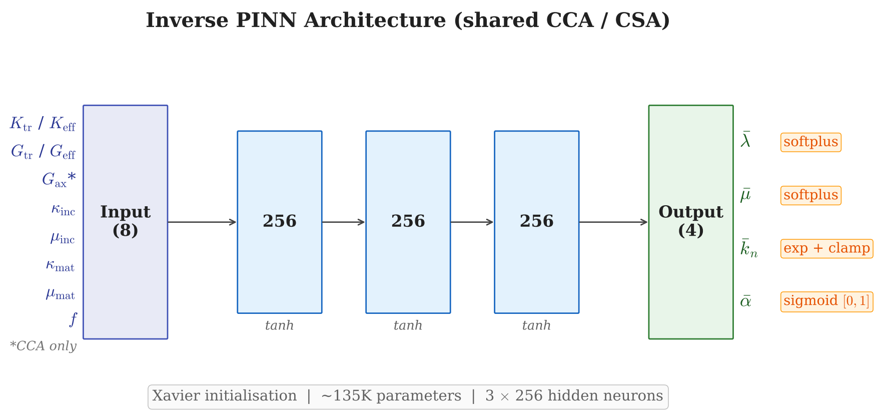

Figure 3.7: Architecture of the inverse PINN shared between CCA and CSA geometries.

Training data generation. We generate 200 000 samples(CCA) by sampling the EGIM parameter space. Material properties are drawn log-uniformly over $\kappa^{(1)} \in [10, 2500]$ and $\mu^{(1)} \in [5, 1500]$ to cover stiffness ratios from 0.05 to 10; log-uniform sampling is necessary because the stiffness ratios spans between 0.05 and 10 and uniform sampling would severely under-represent soft inclusions. Volume fractions are sampled with 30% of points concentrated near the edges of $[0.05, 0.55]$ to reduce prediction errors in extreme configurations. For each sample, the EGIM forward model computes the corresponding effective properties, which become the network inputs. Samples producing non-physical results are discarded.

Loss function. The training loss combines a data term and a physics term. The data term is a mean squared error between predicted and true interface parameters, with $\bar{k}$ scaled down by a factor of 1000 to balance its larger magnitude. The physics term passes the predicted parameters through the EGIM forward model and computes the relative error against the target effective properties, enforcing forward-model self-consistency at every training step. An adaptive weighting scheme inspired by ReLoBRaLo [45] adjusts the relative contribution of each term based on convergence rates, preventing either term from dominating [47]. We train with AdamW [48] ($\eta = 10^{-3}$, weight decay 10^{-4}), cosine annealing [49], batch size 512, and gradient clipping at 1.0 for 500 epochs.

CSA training considerations. The CSA inverse problem requires a modified training strategy because only two effective moduli (K_{eff}, G_{eff}) are available to recover four interface parameters. The training set therefore combines three data sources: 100 000 EGIM-generated samples covering the full parameter space, 5 000 edge samples concentrated near extreme stiffness ratios and volume fractions, and 500 samples derived directly from the 3-layer CSA interphase model. The last source anchors the network to the actual deployment domain, exposing it to the interphase-to-interface mapping during training rather than relying solely on EGIM-generated data. The CSA physics loss penalises the discrepancy against the two target moduli:

$$\mathcal{L} = w_{\text{data}}\,\mathcal{L}_{\text{data}} + w_{\text{phys}}\,\mathcal{L}_{\text{phys}} \tag{3.100}$$

$$\mathcal{L}_{\text{phys}} = \frac{1}{N}\sum_{i=1}^{N}\left[w_K\left(\frac{K_{\text{eff},i}^{\text{pred}} - K_{\text{eff},i}^{\text{target}}}{K_{\text{eff},i}^{\text{target}}}\right)^2 + w_G\left(\frac{G_{\text{eff},i}^{\text{pred}} - G_{\text{eff},i}^{\text{target}}}{G_{\text{eff},i}^{\text{target}}}\right)^2\right] \tag{3.101}$$

with $w_K = 1.5$ and $w_G = 1.0$. The adaptive weights are capped at 150 for w_{phys} and 5 for w_{data} [45]. Training uses AdamW [48] ($\eta = 10^{-3}$, weight decay 10^{-4}), cosine annealing [49] over 500 epochs, batch size 256, and gradient clipping at norm 1.0; the best validation loss is reached at epoch 322.

Algorithms 1 summarise the complete inverse parameter identification pipeline for both the CCA and CSA geometries.

3.3.2 Differential Evolution Baseline

The inverse PINN predicts interface parameters in a single forward pass, but a natural question arises: how close are these predictions to the best parameters achievable by unconstrained optimisation? To answer this, we employ differential evolution (DE) [50] as an independent, per-configuration benchmark. DE is a gradient-free, population-based global optimisation algorithm that iteratively improves a set of candidate solutions through mutation, crossover, and selection. Unlike gradient-based methods, it does not require the objective landscape to be smooth or differentiable, making it well suited for the non-convex parameter spaces encountered in interface model fitting.

For each test configuration, the DE optimiser minimises the sum of squared relative errors between the EGIM-predicted and 3-layer target effective moduli:

$$\min_{\bar{\lambda},\,\bar{\mu},\,\bar{k},\,\bar{\alpha}} \sum_{p}\left(\frac{p^{\text{EI}} - p^{\text{3-layer}}}{p^{\text{3-layer}}}\right)^2 \tag{3.102}$$

where p runs over the relevant effective moduli (K_{eff}, G_{eff} for CSA; K_{tr}, G_{tr}, G_{ax} for CCA). We use a population size of 15, a maximum of 200 iterations, and bound the search space to $\bar{\lambda} \in [0,\,10]$, $\bar{\mu} \in [0.01,\,10]$, $\bar{k} \in [1,\,10\,000]$, and $\bar{\alpha} \in [0,\,1]$.

The key distinction between the two approaches is one of generality versus optimality. The PINN learns a single mapping that covers the entire parameter space

Algorithm 1 Generalized Inverse parameter identification pipeline

Require: Material properties: $\kappa^{(1)}, \mu^{(1)}$ (fibre), $\kappa^{(2)}, \mu^{(2)}$ (matrix); interphase coating properties; radii $r_1, \ldots, r_4$; volume fraction f

Ensure: Interface parameters $\bar{k}, \bar{\lambda}, \bar{\mu}, \bar{\alpha}$ such that EGIM reproduces the interphase effective properties

```
— Solve 3-layer interphase model —
Solve 4 BVPs for CCA and only two BVPs for CSA
Extract T^(q), H^(q) for all q = 1,2,3,4
Compute equivalent tensors: T^eq = Σ c_r T^(r), H^eq = Σ c_r H^(r)
Homogenise via Mori-Tanaka
Effective moduli: K_tr^target, G_tr^target, G_ax^target

— Inverse prediction using neural network —
Assemble input vector: x = [K_tr^target, G_tr^target, G_ax^target, κ^(1), μ^(1), κ^(2), μ^(2), f]
Pass x through the trained inverse PINN → predict k̄, λ̄, μ̄, ᾱ

— Verification using EGIM forward model —
Solve EGIM BVPs using predicted k̄, λ̄, μ̄, ᾱ with same fibre/matrix properties
Homogenise via Mori-Tanaka
Predicted effective moduli: K_tr^pred, G_tr^pred, G_ax^pred
Compute relative errors: e_i = |C_i^pred − C_i^target|/|C_i^target| for each modulus
if all e_i < tolerance then
    return k̄̂, λ̄̂, μ̄̂, ᾱ̂                                  ▷ successful identification
else
    Flag representational gap     ▷ EGIM cannot reproduce interphase behaviour
end if
```

and produces predictions instantaneously, but it must generalise across all configurations. The DE optimiser, by contrast, fits each configuration independently and is therefore free to find the globally best parameters for that specific case. This makes DE an ideal diagnostic tool: if the optimiser also fails to reproduce the target properties, the mismatch must be structural, originating from the Extended Interface model itself rather than from the neural network. In Chapter 4, we use this distinction to separate network prediction errors from fundamental model limitations.

With the inverse network architectures, training strategies, and the differential evolution baseline in place, Chapter 4 presents the results for both geometries: the CSA inverse PINN achieves a 100 % pass rate, while the CCA case reveals a fundamental representational limit of the Extended Interface model that no parameter identification method can overcome.

Chapter 4

Results and Discussion

This chapter presents the numerical results of the framework developed in Chapters 2 and 3. We begin by verifying if the 3-layer interphase model implementation is consistent or not. Section 4.2 then applies the inverse PINN to the CSA geometry and tries to predict the EGIM's parameters that can produce the same Bulk modulus and Shear Modulus(CSA), where the Extended Interface model proves structurally adequate. Section 4.3 does the same study on the CCA geometry: the inverse PINN fails here, and a systematic diagnosis using an differential evolution optimizer reveals that the failure originates not from the neural network but from a fundamental representational limit of the Extended Interface model itself. Section 4.4 synthesises all findings and draws out practical implications.

4.1 Verification of the 3-Layer Interphase Model

Before presenting results that depend on the multi-coated interphase model, we verify the implementation through three independent consistency checks. All tests use the reference material system with $\kappa^{(1)} = 200$ GPa, $\mu^{(1)} = 120$ GPa (fiber) and $\kappa^{(2)} = 20$ GPa, $\mu^{(2)} = 12$ GPa (matrix), at a fibre volume fraction $f = 0.3$.

4.1.1 Concentration Tensor Consistency

The concentration tensors $\mathbb{T}^{(q)}$ and $\mathbb{H}^{(q)}$ for each phase q must satisfy the identity

$$\mathbb{H}^{(q)} = \mathbb{C}^{(q)}\,\mathbb{T}^{(q)}, \tag{4.1}$$

where $\mathbb{C}^{(q)}$ is the stiffness tensor of phase q. We evaluate the identity for all four phases (fiber, three coating layers) and the equivalent inhomogeneity system across three configurations that span the geometric parameter space.

Table 4.1: Maximum absolute error $\max|\mathbb{H} - \mathbb{C}\mathbb{T}|$ across all tensor components, for each phase and three interphase configurations.

Configuration	Fiber	Coat 1	Coat 2	Coat 3	Equivalent
Standard ($\phi = 0.2,\ 0.3,\ 0.4$)	$< 10^{-13}$	$< 10^{-13}$	$< 10^{-13}$	$< 10^{-13}$	$< 10^{-13}$
Wide ($\phi = 0.1,\ 0.5,\ 0.9$)	$< 10^{-13}$	$< 10^{-13}$	$< 10^{-13}$	$< 10^{-13}$	$< 10^{-13}$
Narrow ($\phi = 0.05,\ 0.15,\ 0.25$)	$< 10^{-13}$	$< 10^{-13}$	$< 10^{-13}$	$< 10^{-13}$	$< 10^{-13}$

Table 4.1 confirms that the identity holds to precision of order 10^{-14} across all phases and configurations. This verifies that the four boundary value problems in chapter 3(axial shear, transverse shear, in-plane, and overall axisymmetric) are solved consistently and that the stress concentration tensors are correctly derived from the strain concentration tensors.

4.1.2 Limiting Cases

Two analytical limiting cases provide additional verification.

Fibre limit. When all interpolation weights are set to unity ($w_1 = w_2 = w_3 = 1.0$ in Equation (3.1)), every coating layer takes the inclusion properties and the interphase vanishes. The 3-layer model must then reduce to the standard 2-phase Mori-Tanaka solution. Table 4.2 compares the effective properties from both approaches.

Table 4.2: Fibre limiting case: All coating layers set to fibre properties ($w_1 = w_2 = w_3 = 1.0$).

Property	3-Layer Model	2-Phase MT	Relative Error
K_{tr}	27.9158	27.9158	$< 10^{-12}\,\%$
G_{tr}	16.4756	16.4756	$< 10^{-12}\,\%$
G_{ax}	17.4120	17.4120	$< 10^{-12}\,\%$

Matrix limit. When all phases including the fibre share the matrix properties, the composite must return the pure matrix moduli. This checks that the volume fraction weighting and the homogenization formula (Equations (3.53)–(3.54)) are internally consistent.

Table 4.3: Matrix limiting case: all phases set to matrix properties.

Property	3-Layer Model	Matrix	Relative Error
K_{tr}	20.0000	20.0000	$< 10^{-12}\,\%$
G_{tr}	12.0000	12.0000	$< 10^{-12}\,\%$
G_{ax}	12.0000	12.0000	$< 10^{-12}\,\%$

Both limiting cases are reproduced with precision, which confirms the correctness of the volume fraction computation (Equation (2.18)), the equivalent tensor assembly (Equations (3.53)–(3.54)), and the final homogenization formula.

4.2 CSA: Inverse Parameter Identification via PINN

For the CSA geometry, the spherical BVPs involve only two effective moduli (K_{eff} and G_{eff}) and simpler radial structures than the cylindrical case. We therefore expect the Extended Interface model to be structurally adequate for spherical composites. This section tests that expectation by training the inverse PINN from Section 3.3.1 on CSA data and validating the recovered interface parameters against the 3-layer interphase targets.

4.2.1 Training and Validation

Problem: Given the effective properties K_{eff} and G_{eff} of a multi-coated spherical composite, along with the constituent material properties ($\kappa^{(1)}, \mu^{(1)}, \kappa^{(2)}, \mu^{(2)}$), the volume fraction f, and the inclusion radius r_1, predict the four Extended Interface parameters $\bar{k}$, $\bar{\lambda}$, $\bar{\mu}$, and $\bar{\alpha}$. The network architecture and training strategy follow Section 3.3.1.

Training: We train the CSA inverse PINN on 100,000 samples with constituent properties sampled uniformly: $\kappa^{(1)} \in [0.5, 100]$, $\mu^{(1)} \in [0.5, 50]$ for the particle and $\kappa^{(2)} \in [5, 100]$, $\mu^{(2)} \in [2, 50]$ for the matrix. These ranges are chosen so that the stiffness ratio $\text{SR} = \kappa^{(1)}/\kappa^{(2)}$ spans well below 1 (for soft particle in a stiff matrix) to well above 1 (for stiff particle in a soft matrix), covering the full range of practically relevant composite systems. The combined loss (Equations (3.100) and (3.101)) is minimised over 500 epochs. the full run completes in under one hour on a single CPU core.

Figure 4.1 shows the training loss curves for both geometries. The CSA inverse PINN (left panel) converges smoothly, with all loss components decreasing steadily and the best model selected at epoch 323. The CCA inverse PINN (right panel), also converges but settles at a higher total loss which we can think of as the first indication that the underlying model poses a harder inverse problem for the CCA geometry.

Soft-particle configurations: To validate the trained PINN, we construct five test cases using a soft particle in a stiff matrix: $\kappa^{(1)} = 5$, $\mu^{(1)} = 3$, $\kappa^{(2)} = 20$, $\mu^{(2)} = 12$ (SR $= 0.25$), with $f = 0.3$ and interphase layer thickness ratios $\phi = (0.2, 0.3, 0.4)$. The first three cases use interpolated coating properties that fall between particle and matrix: a weak gradient ($w = 0.10, 0.05, 0.02$), a moderate gradient ($w = 0.75, 0.50, 0.25$), and a strong gradient ($w = 0.95, 0.85, 0.75$). The last two cases test coatings with stiffnesses far beyond both constituents ($\kappa = 100$), which the PINN has never encountered during training.

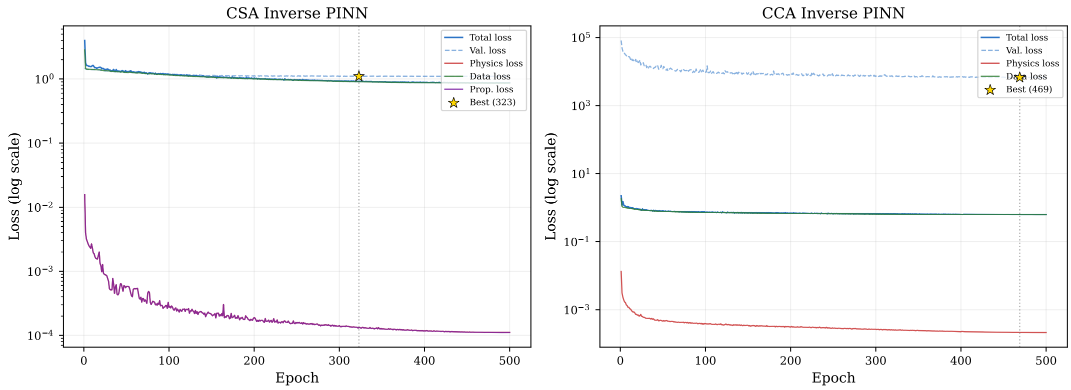

Figure 4.1: Training loss curves for the CSA and CCA inverse PINNs.

If we observe results in Table 4.4, for interpolated coatings (Cases 1-3), the PINN passes two of three cases(threshold 5 %), failing only when the coating properties sit very close to the particle (Case 3). This creates a sharp stiffness jump at the coating–matrix boundary that the zero-thickness Extended Interface model struggles to represent. The out-of-bound cases (Cases 4–5) fail with large errors, particularly for G_{eff}, confirming that the PINN cannot generalise to interphase stiffness profiles outside its training range.

Table 4.4: CSA inverse PINN: soft-particle worked examples (SR = 0.25, $f = 0.3$).

Case	κ: particle $\to$ coatings $\to$ matrix	K_{tgt}	K_{pred}	K_{err} [%]	G_{tgt}	G_{pred}	G_{err} [%]
1	$5 \to 18.5,\ 19.3,\ 19.7 \to 20$	19.651	19.335	1.61	11.957	11.915	0.35
2	$5 \to 8.75,\ 12.5,\ 16.3 \to 20$	17.895	17.881	0.08	11.919	11.354	4.74
3	$5 \to 5.75,\ 7.25,\ 8.75 \to 20$	15.234	15.735	3.29	11.286	10.080	10.68
4	$5 \to 100,\ 10,\ 1 \to 20$	11.687	12.559	7.47	12.286	8.177	33.44
5	$5 \to 1,\ 10,\ 100 \to 20$	25.085	31.028	23.69	21.751	19.398	10.81

Stiff-particle configurations: The five cases in Table 4.4 use a soft particle in a stiff matrix (SR = 0.25). To check whether the PINN generalises to the opposite regime, we repeat the same five configurations of three monotonic gradients plus two out-of-distribution profiles with a stiff particle: $\kappa^{(1)} = 40$, $\mu^{(1)} = 24$, $\kappa^{(2)} = 20$, $\mu^{(2)} = 12$ (SR = 2.0), keeping the same volume fraction and layer thicknesses. Table 4.5 presents the results. The three interpolated cases (1-3) pass comfortably, with the largest error (1.35 % in G_{eff}) well within the 5 % threshold. The out-of-distribution cases (4 and 5) fail, in the pattern observed in Table 4.4.

The two tables gives a clear picture. The Interpolated coating profiles for both soft and stiff particles are reconstructed reliably, with errors typically below 2 %. The PINN struggles when the interphase stiffness profile falls outside the training distribution of Cases 4 and 5 in both Table 4.4 and in Table 4.5 where coating layers

Table 4.5: CSA inverse PINN: stiff-particle worked examples (SR = 2.0, $f = 0.3$).

Case	κ : particle → coatings → matrix	K_{tgt}	K_{pred}	K_{err} [%]	G_{tgt}	G_{pred}	G_{err} [%]
1	$40 \to 22.0,\ 21.0,\ 20.4 \to 20$	20.304	20.287	0.08	12.085	12.087	0.01
2	$40 \to 35.0,\ 30.0,\ 25.0 \to 20$	21.951	21.913	0.17	12.686	12.514	1.35
3	$40 \to 39.0,\ 37.0,\ 35.0 \to 20$	23.667	23.629	0.16	14.083	14.087	0.03
4	$40 \to 100,\ 10,\ 1 \to 20$	11.690	10.188	12.85	1.807	6.528	261.31
5	$40 \to 1,\ 10,\ 100 \to 20$	25.096	26.537	5.74	16.712	16.226	2.91

have stiffnesses far beyond both constituents. Case 4 is particularly severe producing a G_{eff} error exceeding 260 %. For in-distribution configurations, the Extended Interface model is structurally adequate for spherical composites, and the PINN learns an accurate inverse mapping.

Table 4.6 lists the interface parameters that the PINN predicts for each of the ten worked examples. These four numbers ($\bar{k}$, $\bar{\lambda}$, $\bar{\mu}$, $\bar{\alpha}$) are just the network outputs; the reconstruction errors in Tables 4.4 and 4.5 are obtained by feeding them back to the Extended Interface forward model. Two trends are apparent from the predicted values. First, the normal stiffness $\bar{k}$ is considerably larger for the soft-particle cases (∼190–420) than for the stiff-particle cases (∼45–75). When the inclusion is softer than the matrix, the interface must carry more normal stiffness to reproduce the effective bulk response of a finite-thickness graded interphase. Second, the parameter $\bar{\alpha}$ decreases with increasing gradient strength for the soft-particle cases (0.648 → 0.607 → 0.545), suggesting that a sharper property mismatch at the coating–matrix boundary requires a larger traction discontinuity across the interface. The tangential parameters $\bar{\lambda}$ and $\bar{\mu}$ remain in a narrow range across all configurations and do not exhibit a clear systematic trend, which is consistent with the CSA geometry.

Table 4.6: CSA inverse PINN: predicted Extended Interface parameters for all ten worked examples.

Case	Configuration	$\bar{k}$	$\bar{\lambda}$	$\bar{\mu}$	$\bar{\alpha}$
Soft particle (SR = 0.25)					
1	Weak gradient	189.51	3.01	2.55	0.648
2	Moderate gradient	422.65	0.50	3.64	0.607
3	Strong gradient	288.90	1.26	2.92	0.545
4	Out-of-dist. (high→low)	388.36	1.98	3.25	0.409
5	Out-of-dist. (low→high)	318.20	0.05	3.25	0.847
Stiff particle (SR = 2.0)					
A	Weak gradient	45.99	2.23	3.29	0.425
B	Moderate gradient	75.41	3.10	0.98	0.460
C	Strong gradient	64.68	2.54	3.07	0.492
D	Out-of-dist. (high→low)	1.00	3.92	0.01	0.013
E	Out-of-dist. (low→high)	376.73	0.21	3.75	0.555

Broad statistical validation: To assess the generalisation of the trained PINN beyond the hand-picked configurations above, we evaluate it on 500 randomly sampled 3-layer interphase configurations drawn from the training parameter space. Figure 4.2 shows the parity plot comparing EGIM-reconstructed properties against the 3-layer targets, and Figure 4.3 presents the corresponding error distribution.

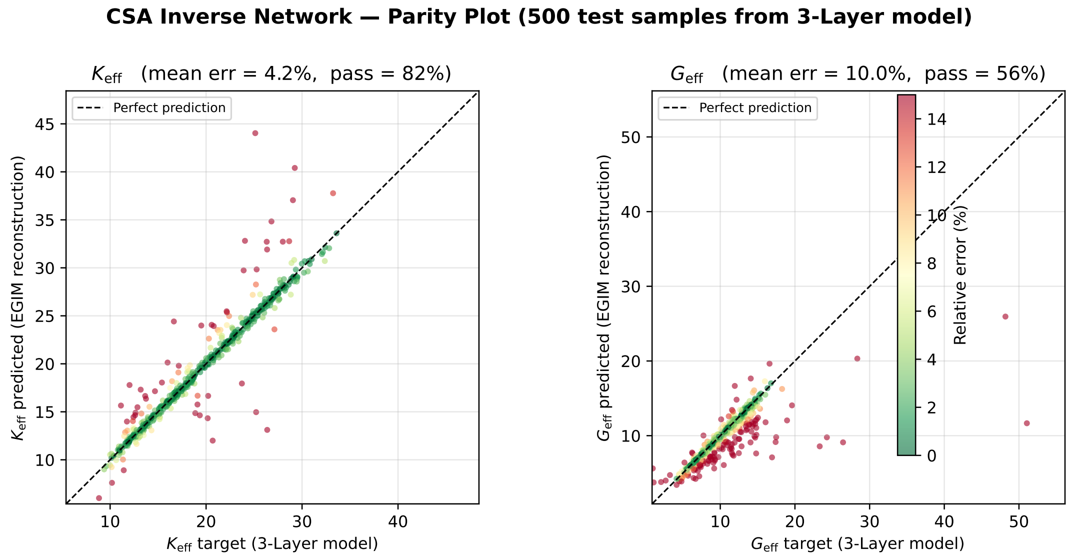

Figure 4.2: CSA inverse PINN: parity plot

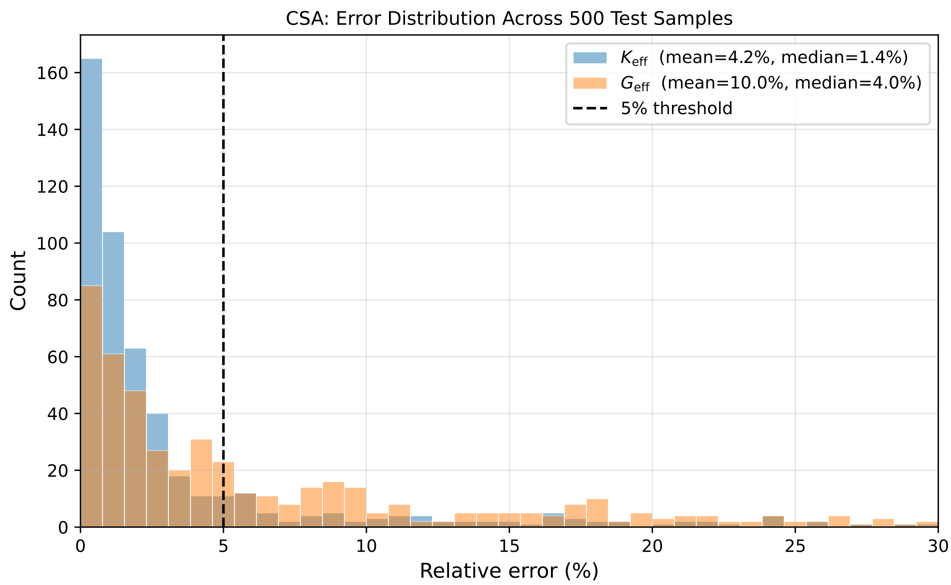

Figure 4.3: CSA error distribution

The bulk modulus reconstruction is reliable across the parameter space, with 82 % of configurations passing the 5 % threshold. The shear modulus is more demanding: its pass rate drops to 56 %. However, the median G_{eff} error of 4.0 % remains below the threshold, indicating that the majority of randomly sampled configurations are predicting well. The remaining outliers probably corresponds to extreme stiffness or unusual interphase profiles near the boundaries of the training domain.

Structured parameter sweep: To visualise how the PINN reconstruction accuracy actually varies, we evaluate the network on a 6×6 grid of $\mathrm{SR} \in \{0.1, 0.25, 0.5, 1.0, 1.5, 2.0\}$ and $f \in \{0.05, 0.10, 0.20, 0.30, 0.40, 0.50\}$, keeping the interphase geometry and grading fixed at the default values ($\phi = (0.2, 0.3, 0.4)$, weights $= (0.75, 0.50, 0.25)$). For each configuration, the 3-layer interphase target is computed, passed through the trained PINN, and the resulting EGIM parameters are forward-evaluated to obtain reconstructed properties.

Figure 4.4 presents the reconstruction errors as heatmaps. Of the 36 configurations, 33 pass the 5 % threshold for both K_{eff} and G_{eff}. The three failures occur at $\mathrm{SR} = 0.1$ with $f \geq 0.30$, where the inclusion is much softer than the matrix and the interphase grading dominates the effective response. Outside this corner, errors remain well below 2 % for both moduli.

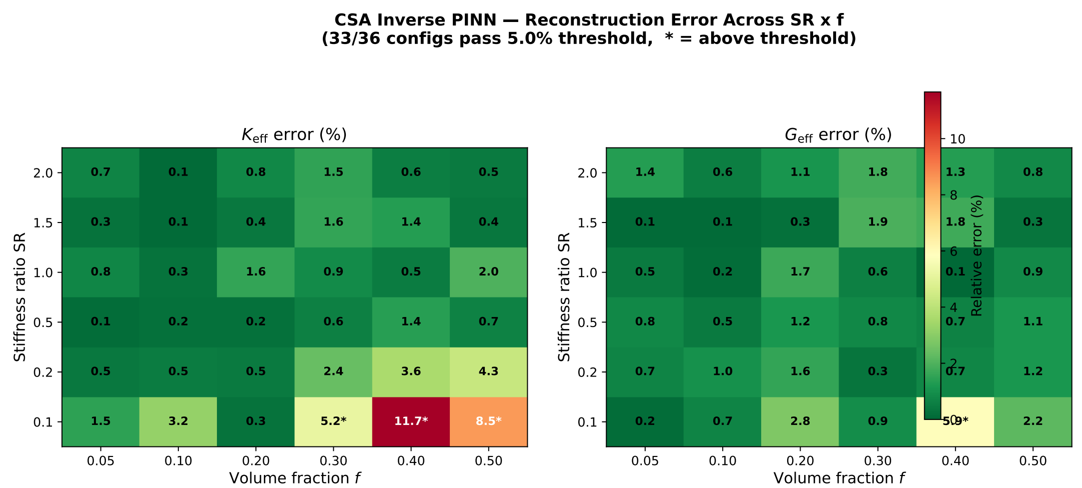

Figure 4.4: CSA inverse PINN error across the SR - f space

4.2.2 CSA Numerical Examples

To further illustrate how the three modelling approaches compare across a continuous parameter range, Figures 4.5–4.7 show sweeps over volume fraction and stiffness ratio for different material configurations.

To check whether this agreement holds across different stiffness contrasts, Figure 4.7 repeats the volume fraction sweep for a soft-particle configuration ($\mathrm{SR} = 0.5$, i.e. $\kappa^{(1)} = 10$, $\mu^{(1)} = 6$, $\kappa^{(2)} = 20$, $\mu^{(2)} = 12$).

Across all three sweeps (Figures 4.5, 4.6, and 4.7), the optimised Extended Interface model overlaps the 3-layer interphase target, confirming that the spherical

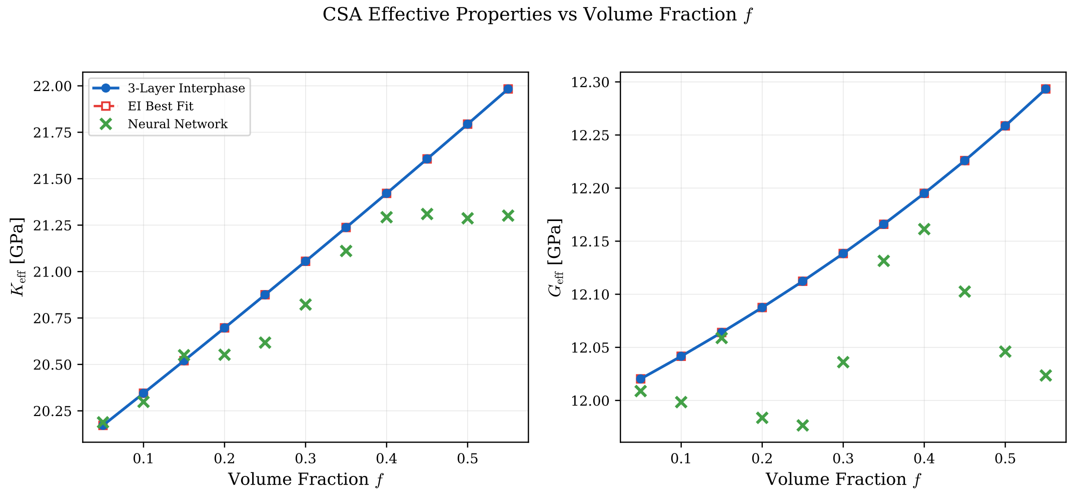

Figure 4.5: CSA effective properties vs. volume fraction at SR = 1.5(stiff particle)

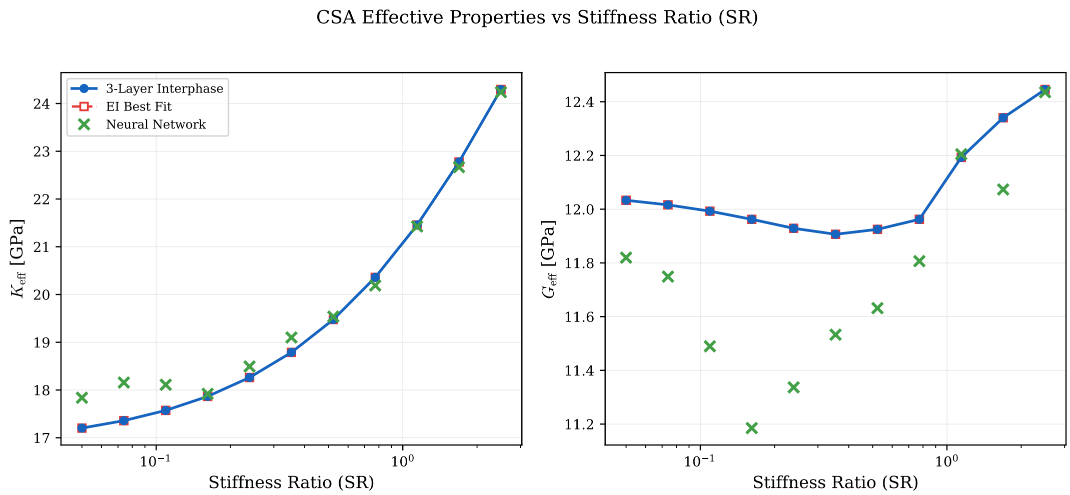

Figure 4.6: CSA effective properties vs. stiffness ratio at $f = 0.3$.

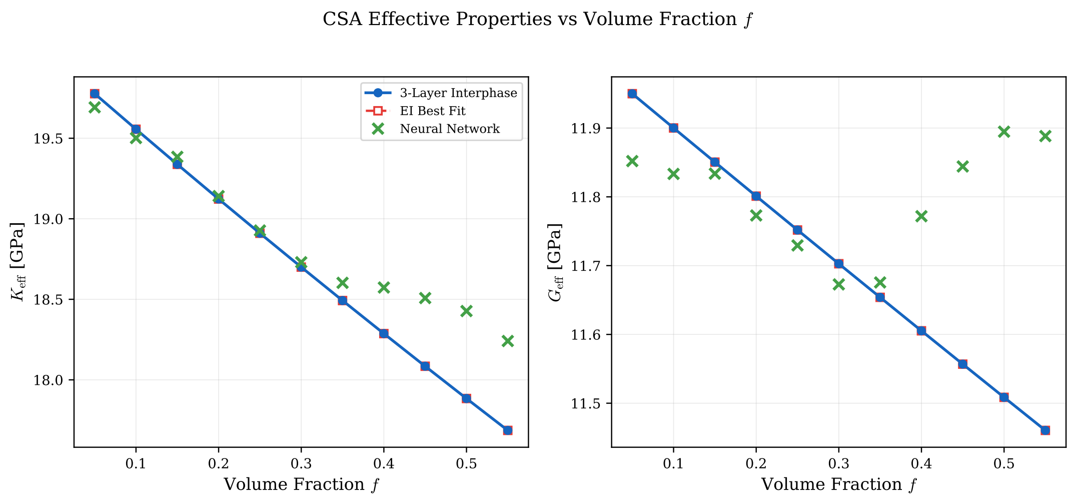

Figure 4.7: CSA effective properties vs. volume fraction at SR = 0.5 (soft particle).

BVPs are well represented by the jump and average conditions of the Extended Interface model regardless of the stiffness contrast. The PINN predictions follow the correct trend for K_{eff} and capture the overall behaviour for G_{eff}, though with more scatter in the shear modulus.

4.3 CCA: Inverse Parameter Identification and Representational Limits

We now apply the same inverse PINN framework to the CCA geometry, where the transversely isotropic BVPs involve three independent effective moduli (K_{tr}, G_{tr}, G_{ax}) and more complex radial displacement patterns. As we shall see, the network fails to reconstruct the 3-layer targets. A systematic diagnosis then reveals that this failure originates not from the neural network but from a fundamental limitation of the Extended Interface model itself.

4.3.1 CCA Inverse PINN Results

The CCA variant of the inverse network takes 8 inputs (K_{tr}, G_{tr}, G_{ax} plus material properties and volume fraction) to produce same four EGIM parameters. We train on 200 000 samples for 500 epochs using the same loss function and adaptive weighting scheme as the CSA, with the training data generated from the CCA Extended Interface forward model. Fibre moduli are sampled log-uniformly over $\kappa^{(1)} \in [10, 2500]$ and $\mu^{(1)} \in [5, 1500]$, while matrix moduli are drawn uniformly from $\kappa^{(2)} \in [10, 50]$ and $\mu^{(2)} \in [5, 25]$. The wider fibre ranges and log-uniform sampling are necessary because fibre-reinforced composites in practice span several orders of magnitude in stiffness contrast, from glass fibres in polymer matrices (SR ≈ 20) to carbon fibres in metal matrices (SR > 50), and uniform sampling over such a range would under-represent the softer end. The volume fraction covers $f \in [0.05, 0.55]$. The full training run takes approximately 2.5 hours on a single CPU core.

Worked examples: To compare directly with the CSA results, we apply the CCA inverse PINN to the same five test cases used in Table 4.4: a soft particle ($\kappa^{(1)} = 5$, $\mu^{(1)} = 3$) in a stiff matrix ($\kappa^{(2)} = 20$, $\mu^{(2)} = 12$), at $f = 0.3$. For each case, we also run a differential evolution optimiser that searches the EI parameter space independently per configuration (population size 15, 200 iterations), establishing a lower bound on the model error.

Table 4.7 presents the results. Neither approach passes any of the five cases. The optimiser achieves lower errors on average. It fits each configuration independently rather than generalising with a single set of weights but even its best-fit solutions

Table 4.7: CCA inverse PINN and differential evolution optimiser: soft fiber

Case	Property	3-Layer Target	PINN Err. (%)	Optim. Err. (%)
1	K_{tr}	19.632	7.2	4.8
	G_{tr}	11.768	30.0	22.1
	G_{ax}	11.798	43.0	8.6
2	K_{tr}	17.833	2.4	4.5
	G_{tr}	10.665	28.4	16.6
	G_{ax}	10.765	35.3	8.1
3	K_{tr}	15.035	10.4	3.2
	G_{tr}	8.905	22.6	6.9
	G_{ax}	9.229	22.9	6.2
4	K_{tr}	11.006	2.4	5.4
	G_{tr}	6.180	0.8	1.7
	G_{ax}	7.304	21.9	5.7
5	K_{tr}	24.415	6.0	2.4
	G_{tr}	14.294	37.8	32.2
	G_{ax}	15.369	24.8	6.1

leave G_{tr} errors above 6 % in four of five cases. Case 4 (non-monotonic, high→low coating stiffness) is the sole exception: both approaches capture G_{tr} within 2 %. However, G_{ax} errors remain above 5 % even here, so the case still fails overall.

In four of the five cases, the PINN drives k_n to similar levels (~ 4900). This parameter saturation observed independently in two different solvers which indicates a structural limitation of the Extended Interface model rather than a learning or optimisation failure. Section 4.3.2 extends this analysis across the full parameter space to map where the two models agree.

Unlike the CSA case, where the PINN fails only for out-of-distribution coating stiffnesses, the CCA failure is structural and persists regardless of the solver or the material configuration.

Stiff-fibre configurations: To verify that the failure pattern is not specific to a soft inclusion in a stiff matrix, we repeat the three monotonic gradient profiles with a stiff fibre: $\kappa^{(1)} = 40$, $\mu^{(1)} = 24$, $\kappa^{(2)} = 20$, $\mu^{(2)} = 12$ (SR = 2.0), keeping $f = 0.3$ and $\phi = (0.2, 0.3, 0.4)$. Table 4.8 presents the results. All three cases fail under both approaches, with G_{tr} and G_{ax} errors well above the 5 % threshold. This confirms that the CCA failure is independent of the stiffness ratio direction, in sharp contrast to the CSA geometry where identical stiff-particle configurations pass comfortably (Table 4.5).

Tables 4.9 and 4.10 list the interface parameters found by the PINN and the differential evolution optimiser, respectively. Unlike the CSA parameters (Table 4.6), which vary meaningfully across configurations, both solvers exhibit characteristic saturation in the CCA case. The PINN drives $\bar{k}$ to values near 5000, while the

Table 4.8: CCA inverse PINN and optimiser: stiff-fiber worked examples (SR = 2.0, $f = 0.3$).

Case	Property	3-Layer Target	PINN Err. (%)	Optim. Err. (%)
A	K_{tr}	20.299	13.1	9.0
	G_{tr}	12.177	14.7	4.1
	G_{ax}	12.185	40.4	10.7
B	K_{tr}	21.914	10.5	7.7
	G_{tr}	13.129	19.8	8.6
	G_{ax}	13.189	34.0	10.1
C	K_{tr}	23.560	8.6	6.6
	G_{tr}	14.082	24.2	13.3
	G_{ax}	14.254	28.2	9.2

optimiser pushes $\bar{k}$ all the way to its upper bound of 10 000 and $\bar{\alpha}$ to 1.0 in seven of eight cases. Meanwhile, the optimiser collapses $\bar{\mu}$ to near-zero values (< 0.06). This consistent saturation pattern across two independent solvers confirms that the Extended Interface model is being stretched to its representational limit, and no reparameterization within the four-parameter isotropic space can close the gap.

Table 4.9: CCA inverse PINN: predicted Extended Interface parameters

Case	Configuration	$\bar{\lambda}$	$\bar{\mu}$	$\bar{k}$	$\bar{\alpha}$
Soft fibre (SR = 0.25)					
1	Weak gradient	2.82	2.66	4591.1	0.430
2	Moderate gradient	2.74	2.16	3275.3	0.343
3	Strong gradient	2.91	1.51	1975.2	0.216
4	Out-of-dist. (high→low)	2.76	0.96	841.1	0.191
5	Out-of-dist. (low→high)	3.08	3.12	4915.8	0.599
Stiff fibre (SR = 2.0)					
A	Weak gradient	2.88	3.18	4883.5	0.386
B	Moderate gradient	2.97	3.37	4915.8	0.425
C	Strong gradient	3.00	3.45	4915.8	0.469

Broad statistical validation: To confirm that this failure pattern extends beyond the five hand-picked configurations, we evaluate the CCA inverse PINN on 500 randomly sampled 3-layer interphase configurations. Figure 4.8 shows the parity plot and Figure 4.9 presents the error distribution.

Across the 500 test configurations, only 2 % of G_{tr} reconstructions fall below the 5 % threshold, with a mean error of 23 %. The K_{tr} pass rate of 45 % and G_{ax} pass rate of 16 % further confirm that the transverse shear gap between the Extended Interface model and the 3-layer interphase persists systematically across the full parameter space.

Table 4.10: CCA DE optimiser: best-fit Extended Interface parameters

Case	Configuration	$\bar{\lambda}$	$\bar{\mu}$	$\bar{k}$	$\bar{\alpha}$
Soft fibre ($SR = 0.25$)					
1	Weak gradient	7.46	0.04	10000.0	1.000
2	Moderate gradient	5.58	0.03	10000.0	1.000
3	Strong gradient	8.11	0.01	10000.0	1.000
4	Out-of-dist. (high→low)	3.06	0.01	898.3	0.000
5	Out-of-dist. (low→high)	1.59	0.11	10000.0	1.000
Stiff fibre ($SR = 2.0$)					
A	Weak gradient	4.89	0.01	10000.0	1.000
B	Moderate gradient	5.39	0.03	10000.0	1.000
C	Strong gradient	2.39	0.05	10000.0	1.000

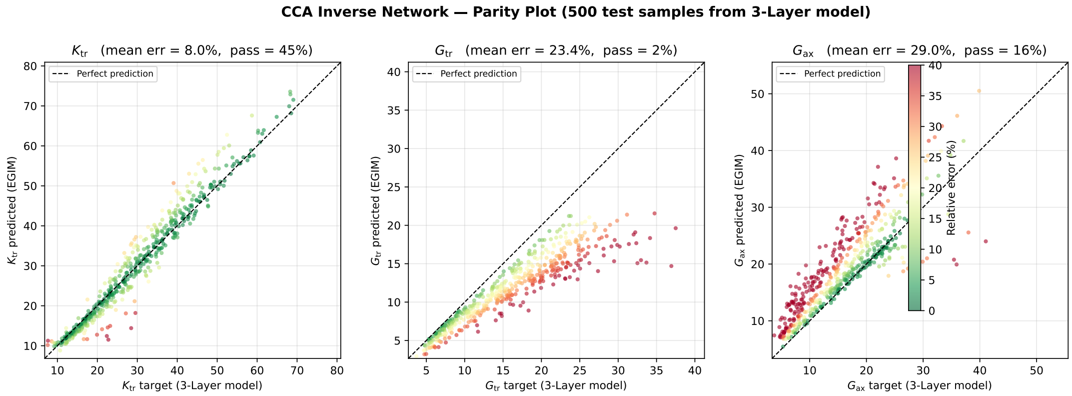

Figure 4.8: CCA inverse PINN: parity plot

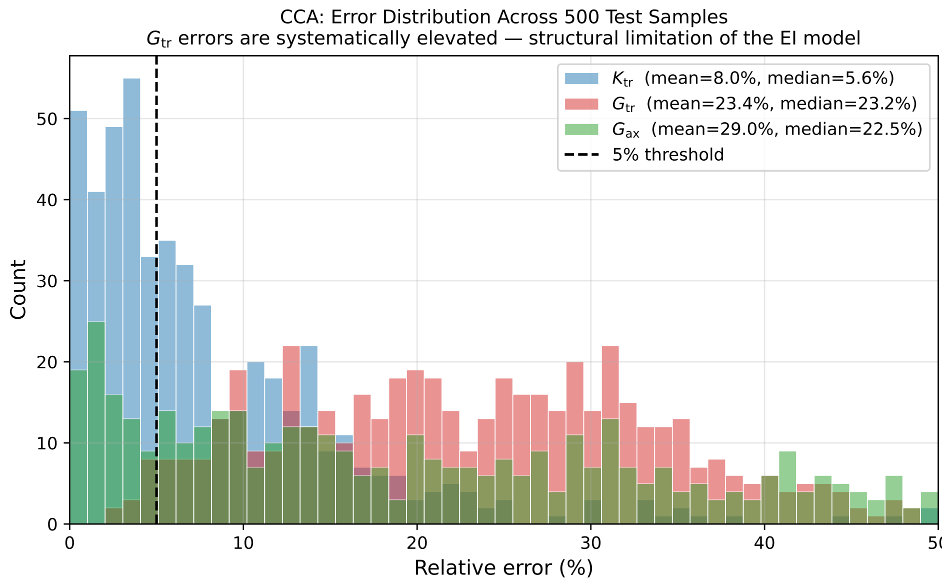

Figure 4.9: CCA error distribution

4.3.2 Diagnosing the Failure: Isotropic Best-Fit (4 Parameters)

The worked examples in Table 4.7 show that even a per-configuration global optimiser cannot close the transverse shear gap. To determine how this gap varies across the full parameter space, we extend the optimisation-based analysis beyond the specific material system of the worked examples.

For each configuration, we solve

$$\min_{\bar{\lambda},\bar{\mu},\bar{k},\bar{\alpha}} \sum_{p\in\{K_{\mathrm{tr}},G_{\mathrm{tr}},G_{\mathrm{ax}}\}} \left(\frac{p^{\mathrm{EI}}-p^{\text{3-layer}}}{p^{\text{3-layer}}}\right)^2 \tag{4.2}$$

using differential evolution (population size 15, 200 iterations) with bounds $\bar{\lambda} \in [0,\ 10]$, $\bar{\mu} \in [0.01,\ 10]$, $\bar{k} \in [1,\ 10\,000]$, and $\bar{\alpha} \in [0,\ 1]$.

Stiffness ratio–volume fraction sweep. To map the region where the two models agree, we conduct a sweep over six stiffness ratios SR $\in \{0.1,\ 0.5,\ 1.0,\ 2.0,\ 5.0,\ 10.0\}$ and six volume fractions $f \in \{0.05,\ 0.10,\ 0.20,\ 0.30,\ 0.40,\ 0.50\}$, giving 36 configurations. Figure 4.10 maps the G_{tr} error across this parameter space. Only 3 of the 36 combinations pass the 5 % threshold, all at SR = 0.1 (soft inclusion) with $f \leq 0.10$. The error grows monotonically with both SR and f, reaching nearly 40 % at SR = 10 and $f = 0.50$. The models agree only where the interphase effect is weak. Figures 4.11 and 4.12 show individual sweeps along the volume fraction and stiffness ratio axes, respectively, with the characteristic G_{tr} gap visible in both.

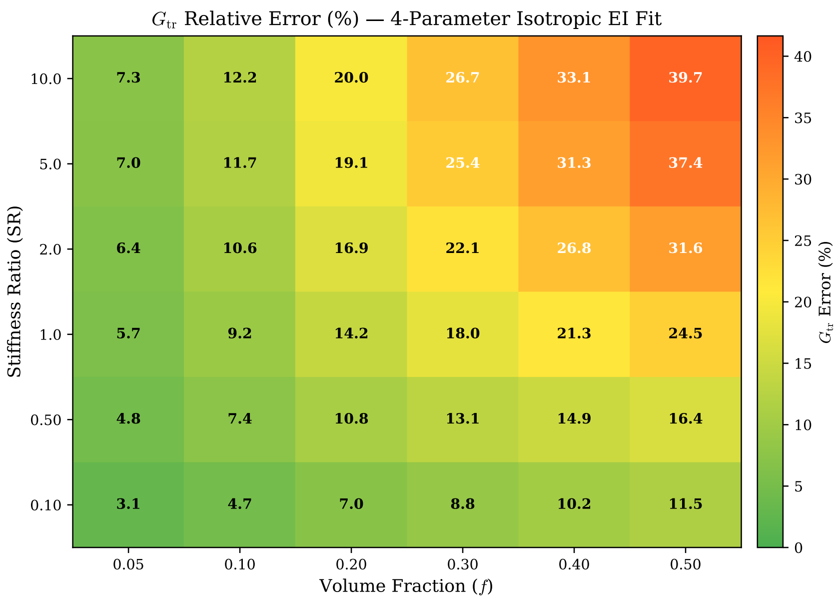

Figure 4.10: G_{tr} relative error (%) between best-fit Extended Interface and 3-layer interphase across the stiffness ratio–volume fraction space.

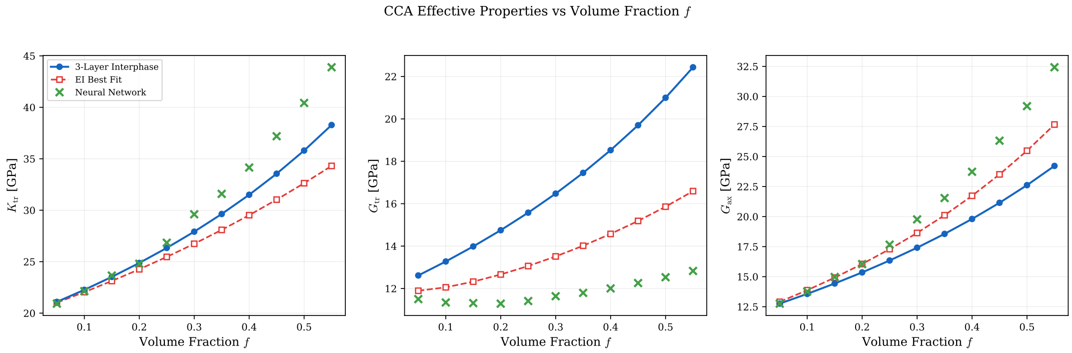

Figure 4.11: CCA effective properties vs. volume fraction at SR = 1.0

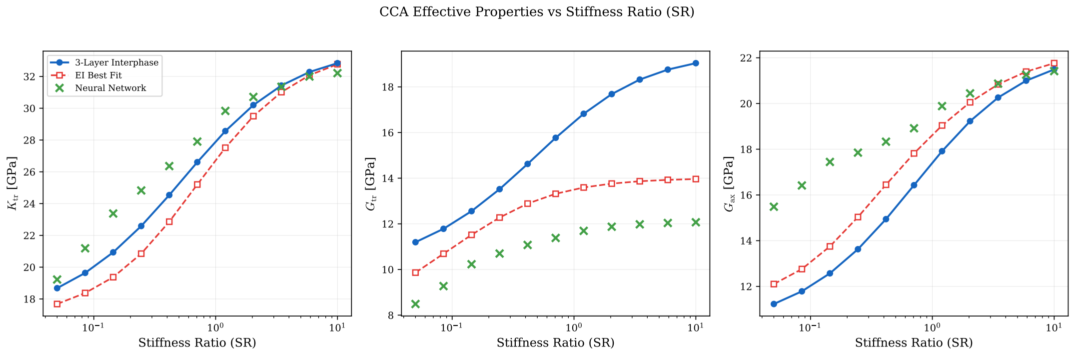

Figure 4.12: CCA effective properties vs. stiffness ratio at $f = 0.3$.

4.3.3 Physical Interpretation

The persistence of the G_{tr} gap points to a fundamental limitation of the zero-thickness approximation. We offer the following physical explanation.

The transverse shear BVP (Section 3.1.2, BVP 2) involves mode-2 mode-4 radial displacement patterns whose amplitudes vary across the coating layers. Within a finite-thickness interphase, each coating layer develops its own stress distribution, and the interaction between mode-2 and mode-4 amplitudes creates a coupling that depends on the radial extent of each layer. The zero-thickness interface collapses this finite thickness response into jump and average conditions at a single surface.

In contrast, the axial shear BVP (BVP 3) and the in-plane bulk BVP (BVP 1) involves a single radial mode. These single-mode problems are well represented by interface conditions. The worked examples support this interpretation: K_{tr} and G_{ax} errors remain moderate in the optimiser column of Table 4.7, whereas G_{tr} is systematically elevated across all configurations.

This finding explains why the inverse PINN fails in Section 4.3.1: the network was trained on data generated from the Extended Interface forward model, and that model is structurally incapable of reproducing the 3-layer transverse shear response. The PINN faithfully learned the EI model but inherited its limitation.

4.4 Discussion

The results of this chapter admit a clear reading once the role of geometry is taken into account.

Why the interface works for CSA. Particle-reinforced composites present the inverse PINN with a tractable problem. The spherical symmetry of the CSA boundary value problems constrains the displacement fields enough that the Extended Interface conditions capture the through-thickness response adequately. The worked examples confirm this for both soft and stiff particles: the recovered parameters $\bar{k}$, $\bar{\lambda}$, $\bar{\mu}$, $\bar{\alpha}$ are physically sensible, and reconstruction errors stay below $2\,\%$ for interpolated coating cases. The one failed case involves an out-of-distribution coating stiffness a distributional issue, not a model one. For CSA geometries, the inverse PINN gives a reliable mapping from effective properties to interpretable interface parameters.

Why the interface fails for CCA. Fibre-reinforced composites are a different matter. As discussed in Section 4.3.3, the shear BVP couples mode-2 and mode-4 radial displacement patterns across the finite-thickness coating layers, and the zero-thickness interface cannot encode this finite thickness mode interaction.. The consequence is the G_{tr} gap that no parameter tuning can eliminate.

Two independent lines of evidence support this conclusion. The per-configuration optimiser and the inverse PINN, despite their quite different properties, agree on the failure pattern across nearly all tested configurations. The stiffness ratio–volume fraction sweep (Figure 4.10) makes the scope of this limitation plain: the interface model passes the $5\,\%$ threshold only at low stiffness contrast and low volume fraction precisely the regime where the interphase effect is itself small and an interface approximation is least needed.

Practical implications. The asymmetry between the two geometries has a direct consequence for how this framework should be used in practice. For particle composites, the PINN-recovered parameters can be fed into the analytical forward model in a single evaluation the approach works and the recovered parameters carry physical content. For fibre composites, any route through the Extended Interface model will carry a G_{tr} error that no amount of parameter tuning can eliminate; the better option is to work with the 3-layer interphase model directly. As a practical diagnostic for any new composite system, the optimisation-based gap check of Section 4.3.2 can serve as a first test before committing to the interface route.

Chapter 5

Conclusion

5.1 Summary of Contributions

This thesis has developed and evaluated a computational framework for predicting the effective properties of composites with graded interphases, combining analytical micromechanics with neural network approaches. The main contributions are:

1. **Verified 3-layer interphase model.** This study has implemented and verified the multi-coated CCA and CSA interphase models, solving four boundary value problems for the CCA geometry and two for the CSA geometry. The implementation passes all consistency checks: $\mathbb{H} = \mathbb{C}\,\mathbb{T}$ to machine precision ($\sim 10^{-14}$), both limiting cases reproduce the expected 2-phase results, and volume fractions sum to unity (Section 4.1).

2. **Inverse PINN for CSA composites.** A physics-informed neural network (PINN) was successfully developed to recover Extended Interface parameters given the effective material properties of Three layered Interphase model. For interpolated coating profiles covering both soft and stiff particles all tested configurations pass the 5 % threshold with errors typically below 2 %. Out-of-distribution coatings, with stiffnesses far beyond both constituents, are not reliably recovered; this is a data-coverage limitation, not a structural one (Section 4.2.1).

3. **Independent confirmation via CCA inverse PINN.** The same inverse Neural Network is applied to CCA composites. On the same five test cases used for CSA, neither the PINN nor a per-configuration global optimiser passes any configuration. G_{tr} errors exceed 20 % for the PINN and 6 % for the optimiser in four of five cases; Case 4, whose compliant outer layer weakens the mode coupling, is the sole exception and is consistent with the physical explanation of Section 4.3.3. This confirms the gap analysis from a different angle: the limitation is structural, not an artifact of the optimisation procedure (Section 4.3.1).

4. **Quantification of the interface–interphase gap.** The Extended Interface model couldnot reproduce the transverse shear modulus of a CCA multi-layer interphase Fig 4.11. A stiffness ratio–volume fraction sweep reveals that only 3

of 36 parameter combinations meet the 5 % acceptance threshold, with the gap growing monotonically with stiffness contrast and volume fraction. This establishes a fundamental limitation of the zero-thickness approximation for fibre composites (Section 4.3).

5.2 Limitations

The framework has several limitations that should be considered when interpreting the results.

The inverse PINN approach is limited to CSA composites. For CCA composites, the Extended Interface model cannot represent the transverse shear response, making inverse parameter identification through that model structurally impossible. Alternative interface theories that retain both volumetric and deviatoric information could potentially bridge this gap, but such theories are beyond the scope of this work.

The neural network depends on the quality and coverage of the training data. The training set is generated from the EGIM forward model, and the network's ability to generalise depends on how well the EGIM parameter ranges cover the physically relevant region. In practice, any deployment should include a check that the query point falls within the training distribution.

All results assume linear elastic constituent behaviour, perfect bonding between phases, and spherical or cylindrical inclusion geometries. Extensions to nonlinear materials, imperfect interfaces, or non-canonical geometries would require new training datasets and potentially modified network architectures.

5.3 Future Work

Several directions for future research emerge from this work:

- **Higher-order interface theories for CCA.** The G_{tr} gap identified in this thesis calls for interface models that retain the transverse shear response of the interphase. Second-gradient or micromorphic interface theories could potentially capture the transverse–axial shear coupling that the zero-thickness approximation misses.

- **Extension to nonlinear and viscoelastic constituents.** The current framework assumes linear elastic behaviour. Many interphase materials (e.g. polymer coatings, oxide layers) exhibit rate-dependent or nonlinear responses that would require reformulating the boundary value problems and retraining the neural networks.

- **Integration with finite element methods.** The interface parameters recovered by the inverse PINN could serve as inputs to constitutive models within multiscale FEM simulations, providing effective properties at each integration point without solving the micro-scale BVPs. This would enable large-scale parametric studies of composite structures with graded interphases.

- **Experimental validation.** Comparing the predicted effective properties against experimental measurements from nanoindentation or resonant ultrasound spectroscopy would establish the practical accuracy of the framework beyond the analytical benchmarks used in this thesis.

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